Understanding Inverse Reinforcement Learning under Overparameterization: Non-Asymptotic Analysis and Global Optimality

Ruijia Zhang
School of Science and Engineering
The Chinese University of Hong Kong, Shenzhen
ruijiazhang@link.cuhk.edu.cn

Siliang Zeng
Department of Electric and Computer Engineering
University of Minnesota
zeng0176@umn.edu

Chenliang Li
Department of Industrial Engineering
Texas A&M
chenliangli@tamu.edu

Mingyi Hong
Department of Electric and Computer Engineering
University of Minnesota
mhong@umn.edu

Alfredo Garica
Department of Industrial Engineering
Texas A&M
alfredo.garcia@exchange.tamu.edu

Abstract

The goal of Inverse reinforcement learning (IRL) task is to identify the underlying reward function and the corresponding optimal policy from a set of expert demonstrations. Most algorithms in IRL assume the reward has a linear structure. In this work, we want to extend our understanding of the IRL problem when the reward is parametrized by some neural network structures. We first propose a novel algorithm to effectively solve the neural IRL problem. Meanwhile, conventional IRL algorithms usually adopt a nested structure, thus exhibiting computational inefficiency, especially when the MDP is high dimensional. We address this problem by proposing the first authentic neural single-loop maximum likelihood algorithm. Due to the nonlinearity of reward function neural network approximation, the previous convergence result is no longer guaranteed. We give the nonasymptotic convergence analysis of our proposed algorithm by using the overparametrization of certain neural networks. However, it remains mysterious whether our algorithm can identify the globally optimal reward and the corresponding optimal policy, Under some overparameterized neural network parameterization, we provide positive answers to both questions. To our knowledge, this is the first IRL algorithm with a finite-time convergence guarantee that identifies global optimum with neural network parameterization.

Preprint. Under review.

1 Introduction

Given the observed trajectories of states and actions executed by an expert, we consider the challenge of inferring the reinforcement learning reward function in which the expert was trained. This problem is commonly known as inverse reinforcement learning (IRL), as discussed in a recent survey by Osa et al. (2018). IRL involves the task of estimating both the reward function and the expert's policy that best aligns with the provided data. While there are certain limitations on the identifiability of rewards Kim et al. (2021), the process of estimating rewards from expert trajectories offers valuable insights for estimating optimal policies under different environment dynamics and/or learning new tasks through reinforcement learning.

In the Inverse Reinforcement Learning (IRL) formulation introduced by Ziebart et al. (2008), the model representing the expert's behavior is defined as the policy that maximizes entropy while simultaneously adhering to a constraint that ensures the expected features. Under such a policy, the expected features match the empirical averages observed in the expert's dataset. The algorithms developed for MaxEnt-IRL Ziebart et al. (2008, 2010); Wulfmeier et al. (2015) employ a nested loop structure. This structure alternates between an outer loop involving a reward update step and an inner loop responsible for computing explicit policy estimates. While this nested structure is computationally manageable in tabular environments, it becomes computationally demanding in high-dimensional settings that require function approximation.

Most recently, a new formulation of IRL based on maximum likelihood (ML) estimation has been proposed. Zeng et al. (2022b) It has been established that a robust duality relationship exists in the case of maximum entropy Inverse Reinforcement Learning (IRL) when rewards are expressed as a linear combination of features. Zeng et al. (2022a) The ML formulation is a bi-level optimization problem. In this framework, the upper-level problem is concerned with maximizing the likelihood function, while the lower-level problem seeks to identify the optimal policy given the current reward parameterization. To alleviate the computational burden associated with the recurrent optimization of the lower-level policy, ML-IRL Zeng et al. (2022b) carries out the policy improvement step and reward optimization step alternately. This approach ensures that each step can be executed with relatively lower computational costs. However, whether ML-IRL converges or not relies on policy improvement step Zeng et al. (2023), which has not been thoroughly discussed before.

The new policy at each round is generated by *soft policy iteration* Haarnoja et al. (2017). ML-IRL Zeng et al. (2023) assumes it has an $\epsilon-$accurate estimation of soft Q function, which is essential for controlling global convergence. Hence, the effectiveness of the single-loop ML-IRL structure hinges on obtaining precise policy estimations, a critical aspect that remains intractable in practice. Reported results in the Zeng et al. (2022b) are derived from a single-loop implementation. The policy evaluation is carried out using multiple iterations of soft SAC. It's worth noting that this approach cannot consistently provide $\epsilon-$accurate estimations for each policy update. The paper therefore lacks a convergence guarantee to substantiate its performance.

Usually, policy evaluation is done by following the temporal-different learning (TD) paradigm Sutton (1988). In particular, in deep reinforcement learning, TD is often applied to learn value functions parametrized by neural networks Lillicrap et al. (2015); Mnih et al. (2016); Haarnoja et al. (2018), which gives rise to neural TD. Neural TD does not follow the stochastic gradients of any objective function, hence leading to possible divergence, which makes the convergence analysis challenging. Recently, nonasymptotic global convergence of neural TD and its several variants has been given in Cai et al. (2023). However, neural TD requires infinite iterations to provide an $\epsilon-$accurate policy estimation. It's not computationally feasible for an IRL algorithm to run sufficiently many rounds of policy evaluations before achieving a policy update.

To tackle this problem, we design a single-loop ML-IRL framework. Similar to Zeng et al. (2022b), we improve the policy and reward update step alternately. The two-timescale stochastic approximation (TTSA) approach Hong et al. (2020); Borkar (1997) is also adopted for stability consideration. Besides, each policy improvement step is followed by only a single iteration of soft Q-learning. We carefully select the different stepsizes of the reward update step and soft Q-learning to ensure the global convergence of our algorithm. The convergence analysis draws insights from literature related to two-timescale single-loop algorithms. By separating actor and critic in an asymptotic sense, Fu et al. (2021) gives the global convergence analysis of actor-critic where both actor and critic function are approximated by linear or deep neural networks.

Moreover, existing ML-IRL works only consider the optimality analysis when the reward has a linear structure. Zeng et al. (2022b, 2023) We want to advance IRL research by understanding IRL under neural network parametrization. A series of works show that neural networks with wide layers can approximate arbitrary continuous functions. Telgarsky (2015, 2016); Lu et al. (2017); Liang and Srikant (2016); Yarotsky (2017, 2018); Hanin and Sellke (2017); Hanin (2019). Specifically, a double-layer neural network with ReLU activation is usually considered for theoretical analysis. Hofmann et al. (2008) showed that it can represent a rich function class. Neural network parametrized objectives appear to be nonconvex and nonsmooth, which made the theoretical analysis difficult. Related works on the optimality analysis of neural networks such as Li and Liang (2018); Du and Lee (2018) showed that for a one-hidden-layer network with ReLU activation function using overparameterization and random initialization, GD and SGD can find the near global-optimal solutions in polynomial time. Zou et al. (2018) extended the result to L-hidden-layer-fully-connected neural network structure. Several works Du et al. (2018); Cai et al. (2023); Gaur et al. (2023) discussed optimality results in deep reinforcement learning. However, previous results cannot be easily applied to the bi-level optimization formulation. In this work, we make the first attempt to show that our single-loop IRL algorithm can provably converge to global optimal results under overparametrization.

Challenges: The main challenge is how to ensure the global convergence of our IRL algorithm with only partial policy improvement, due to imprecise policy evaluation. Besides, the strong duality result Zeng et al. (2022b) fails to hold since the reward is now parametrized by neural networks. What's more, the highly nonconvex bi-level structure brings a lot of difficulties in optimality analysis.

Our Contributions: We design the first authentic single-loop maximum likelihood IRL algorithm capable of producing high-quality estimates of both rewards and behavior policies. We also extend the linear reward setting to a neural network parametrization. A thorough theoretical understanding of the convergence and optimality guarantees is provided by studying the local linearization property of our neural network parametrization. The major contributions of this work are listed below.

- We first bridge the gap in the previous policy evaluation step of ML-IRLZeng et al. (2022b) by proposing a nested algorithm called dynamically truncated ML-IRL. We incorporate a variant of the soft Q-learning algorithm for policy estimation, which is followed by *soft policy iteration*. Haarnoja et al. (2017) The number of iterations needed in soft Q-learning increases linearly as the reward parameter updates. To reduce computation complexity aroused by nested structure, we then design an authentic single-loop ML-IRL algorithm to solve the bi-level IRL problem. We carefully control the stepsizes of TD update in soft Q-learning, so that it only needs one single iteration of soft Q-learning for policy estimation.

- Further, we show that our single-loop algorithm has strong theoretical guarantees: to achieve a certain ϵ-approximate global solution for a neural network over-parametrized parameterized problem, it requires $\mathcal{O}\left(\epsilon^{-8}\right)$ steps of policy and reward updates each. To our knowledge, it is the first algorithm that has a finite-time convergence analysis for the IRL problem under neural network parameterization of reward functions.

- Notably, We develop the approximate global optimality analysis of the solution our algorithm returns under overparametrization. Previous ML-IRL algorithms are only able to find the global optimum in cases where the reward is linearly parameterized Zeng et al. (2022b). However, we extend the IRL problem setting to include general neural network parameterized rewards and show that, with certain overparametrized neural network structures, it is still possible to find the global optimum. This is the first neural ML-IRL algorithm with global optimality guarantees.

2 Preliminaries

In RL, a discounted Markov decision process (MDP) is denoted by a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, d, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P(\cdot \mid s, a)$ is the transition probability kernel, $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function, $d(\cdot)$ is the initial state distribution and γ is the discount factor. A policy π takes state $s \in \mathcal{S}$ as an input and gives a distribution over actions $\mathcal{A}$.

2.1 Maximum Entropy Reinforcement Learning and Soft Value Functions

Maximum entropy RL Haarnoja et al. (2017) searches the optimal policy, which maximizes its entropy at each visited state:

$$\pi^*_{\text{MaxEnt}} = \arg\max_{\pi} \sum_t \mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\mu_\pi} \left[\gamma^t \left(r\left(\mathbf{s}_t,\mathbf{a}_t\right) + \mathcal{H}\left(\pi\left(\cdot \mid \mathbf{s}_t\right)\right)\right)\right], \tag{1}$$

In this context, the soft V-function and soft Q-function are defined as follows:

$$V^{\text{soft}}_{r_k,\pi_k}(s) = \mathbb{E}_{\pi_k, s_0=s}\left[\sum_{t=0}^{\infty} \gamma^t \left(r\left(s_t, a_t; \theta_k\right) + \mathcal{H}\left(\pi_k\left(\cdot \mid s_t\right)\right)\right)\right]$$

$$Q^{\text{soft}}_{r_k,\pi_k}(s,a) = r\left(s,a;\theta_k\right) + \gamma \mathbb{E}_{s'\sim\mathcal{P}(\cdot|s,a)}\left[V^{\text{soft}}_{r_k,\pi_k}(s)\right]$$

The optimal entropy-regularized policy of equation (1) at $k+1$th visited state can be expressed as $\pi_{k+1}(\cdot \mid s) \propto \exp\left(Q^{soft}_{r,\pi_k}(s,\cdot)\right), \forall s \in \mathcal{S}$ Ziebart et al. (2010); Haarnoja et al. (2017). Under a fixed reward function, it can be shown that the new policy π_{k+1} monotonically improves π_k, and it converges linearly to the optimal policy; see Theorem 4 in Haarnoja et al. (2017) and Theorem 1 in Cen et al. (2022).

2.2 Maximum Entropy Inverse Reinforcement Learning (MaxEnt-IRL)

Given a dataset $\mathcal{D}$ which contains trajectories $\tau = \{(s_t, a_t)\}_{t=0}^{\infty}$ sampled from an expert policy, the MaxEnt-IRL formulation Ziebart et al. (2010, 2013); Bloem and Bambos (2014) consists of finding a policy maximizing entropy subject to the expected features under such policy matching the empirical averages in the expert's observation dataset. Specifically, the MaxEnt-IRL formulation is given by:

$$\max_{\pi} H(\pi) := \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{\infty} -\gamma^t \log \pi\left(a_t \mid s_t\right)\right]$$

$$\text{s.t. } \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{\infty} \gamma^t \phi\left(s_t, a_t\right)\right] = \mathbb{E}_{\tau\sim\mathcal{D}}\left[\sum_{t=0}^{\infty} \gamma^t \phi\left(s_t, a_t\right)\right]$$

where $\phi(s,a)$ is the feature vector of the state-action pair (s,a).

MaxEnt-IRL has been discussed theoretically Bloem and Bambos (2014); Zeng et al. (2022a,b, 2023) and applied in practice Zhou and Small (2020); Fu et al. (2019); Wu et al. (2020) before. From a computational perspective, it needs to repeatedly solve optimal π^*_θ under each reward function r_θ. Such subroutine can be intractable, particularly when the underlying MDP is high dimensional.

2.3 Problem Formulation: Maximum Log-Likelihood IRL (ML-IRL)

In this section, we review the fundamentals of maximum likelihood inverse reinforcement learning (ML-IRL).

Let $\mathcal{D} := \{\tau_i\}_{i=1}^N$ denote the dataset containing observed trajectories. A model of the expert's behavior is a randomized policy $\pi_\theta(\cdot \mid s)$ where θ is a parameter vector. Assuming the state dynamics $\mathcal{P}\left(s_{t+1} \mid s_t, a_t\right)$ are known, the discounted log-likelihood of observing a sample trajectory τ under model π_θ can be written as follows:

$$\mathbb{E}_{\tau\sim\mathcal{D}}\left[\log \prod_{t\geq 0} \left(\mathcal{P}\left(s_{t+1} \mid s_t, a_t\right) \pi_\theta\left(a_t \mid s_t\right)\right)^{\gamma^t}\right]$$

$$= \mathbb{E}_{\tau\sim\mathcal{D}}\left[\sum_{t\geq 0} \gamma^t \left(\log \pi_\theta\left(a_t \mid s_t\right) + \log \mathcal{P}\left(s_{t+1} \mid s_t, a_t\right)\right)\right].$$

Let

$$L(\theta) := \mathbb{E}_{\tau\sim\mathcal{D}}\left[\sum_{t>0} \gamma^t \log \pi_\theta\left(a_t \mid s_t\right)\right] \tag{2}$$

Then the IRL problem can be formulated as the following maximum log-likelihood IRL formulation Zeng et al. (2022b):

$$\begin{aligned} &\max_{\theta} \; L(\theta) \\ &\text{s.t} \;\; \pi_\theta := \arg\max_{\pi} \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t \left(r\left(s_t, a_t; \theta\right) + \mathcal{H}\left(\pi\left(\cdot \mid s_t\right)\right)\right)\right] \end{aligned} \tag{3}$$

where $r(s, a; \theta)$ is the reward function and $\mathcal{H}(\pi(\cdot \mid s)) := -\sum_{a \in \mathcal{A}} \pi(a \mid s) \log \pi(a \mid s)$ denotes the entropy of policy $\pi(\cdot \mid s)$. Note that the problem takes the form of a bi-level optimization problem, where the upper-level problem (ML-IRL) optimizes the reward parameter θ, while the lower-level problem describes the expert's policy as the solution to an entropy-regularized MDP Haarnoja et al. (2017, 2018). Note that the entropy regularization in equation (3) ensures the uniqueness of the optimal policy π_θ given the fixed reward function $r(s, a; \theta)$ Haarnoja et al. (2017); Cen et al. (2022). Even when the underlying MDP is high-dimensional and complex, the optimal policy could still be obtained; see recent developments in Haarnoja et al. (2017, 2018).

3 Proposed Algorithm

3.1 Neural Network Parametrization of IRL

We consider $\mathcal{S}$ to be continuous and $\mathcal{A}$ to be finite. We represent the state-action pair $(s, a) \in \mathcal{S} \times \mathcal{A}$ by a vector $x = \psi(s, a) \in \mathcal{X} \subseteq \mathbb{R}^d$ with $d > 2$, where ψ is a given one-to-one feature map. With a slight abuse of notation, we use (s, a) and x interchangeably. Without loss of generality, we assume that $\|x\|_2 = 1$ and $|r(x)|$ is upper bounded by a constant $\bar{r} > 0$ for any $x \in \mathcal{X}$.

To parametrize soft Q values and reward function, we use a two-layer neural network

$$\hat{Q}(x; W) = \frac{1}{\sqrt{m}} \sum_{r=1}^{m} b_r \sigma\left(W_r^\top x\right) \tag{4}$$

$$\widehat{r}(x; \theta) = \frac{1}{\sqrt{m}} \sum_{r=1}^{m} b_r \sigma\left(\theta_r^\top x\right) \tag{5}$$

Here σ is the rectified linear unit (ReLU) activation function $\sigma(y) = \max\{0, y\}$. $b = (b_1, \ldots, b_m)$ are initialized as $b_r \sim \mathrm{Unif}(\{-1, 1\})$, the parameter W is initialized as $W_r \sim N\left(0, I_d/d\right)$, and the parameter θ is initialized as $(\theta_1, \ldots, \theta_m)$ $\theta_r \sim N\left(0, I_d/d\right)$ for any $r \in [m]$ independently. During training, we only update $W = (W_1, \ldots, W_m), \theta = (\theta_1, \ldots, \theta_m) \in \mathbb{R}^{md}$, while keeping $b = (b_1, \ldots, b_m) \in \mathbb{R}^m$ fixed as the random initialization. This is a typical network structure for theoretical analysis, due to its local linearization property Wang et al. (2019). It was shown to be able to learn a class of infinite-order smooth functions Arora et al. (2019); Allen-Zhu et al. (2020).

To solve problem (3), Zeng et al. (2022b) proposed an algorithm named ML-IRL based on certain (stochastic) bi-level optimization algorithms Hong et al. (2020); Ji et al. (2021); Khanduri et al. (2021), which to some extent reduces the high complexity arising from the traditional nested loop algorithms. Specifically, *soft policy iteration* Haarnoja et al. (2017) is adopted to optimize the lower-level problem. Therefore, approximating soft Q-function is critical for policy improvement in ML-IRL. However, one key drawback of ML-IRL is that we normally do not have direct access to the exact soft Q-function. In order to bridge the gap, we decide to use a few stochastic update steps in soft Q-learning Haarnoja et al. (2017) to perform the policy evaluation.

We then present Algorithm 1 Haarnoja et al. (2017)to demonstrate such an alternative implementation of policy estimation in ML-IRL Zeng et al. (2022b).

3.2 Maximum Likelihood Inverse Reinforcement Learning (ML-IRL) with Dynamically truncated Soft Q-learning nested

Typically, it requires $\mathcal{O}\left(\epsilon^{-2}\right)$ loops of Algorithm 1 to get an $\epsilon-$accurate soft Q-function for each policy improvement. Cai et al. (2023) However, it is not impossible to do so in real practice. Therefore,

Algorithm 1 Policy Evaluation (T): Neural Soft Q-learning

Input:

for $t = 0, 1, \ldots, T-1$ **do**

  Sample a tuple (s, a, r, s') from the stationary distribution μ_k of the exploration policy π_k

  Bellman residual calculation: $\delta \leftarrow \widehat{Q}(s, a; W(t)) - r - \gamma \operatorname{softmax}_{a' \in \mathcal{A}} \widehat{Q}(s', a'; W(t))$;

  TD update: $\widetilde{W}(t+1) \leftarrow W(t) - \eta \delta \cdot \nabla_W \widehat{Q}(s, a; W(t))$;

  Projection: $W(t+1) \leftarrow \operatorname{argmin}_{W \in S_B} \|W - \widetilde{W}(t+1)\|_2$

  Averaging: $\bar{W} \leftarrow \frac{t+1}{t+2} \cdot \bar{W} + \frac{1}{t+2} \cdot W(1)$

end for

Output: $Q^{soft}_{r_{\theta_k}, \pi_k}(s, \cdot) = \widehat{Q}_{\text{out}}(\cdot) \leftarrow \widehat{Q}(\cdot; \bar{W})$

we now give our Algorithm 2. The innovation in Algorithm 2 lies in the dynamic number of stochastic steps in Algorithm 1. At the Kth round of policy improvement, it requires $K+2$ iterations of Algorithm 1 to get the approximate soft Q value needed. This design makes the soft Q-learning get a policy estimation in finite iterations. However, the previous convergence guarantee is not necessarily guaranteed now, because it doesn't arrive at an $\epsilon-$accurate policy estimation as assumed in ML-IRL Zeng et al. (2023), especially in the early rounds. We'll give our theoretical results in the section 4. We now present the comprehensive details in Algorithm 2.

Algorithm 2 Maximum Likelihood Inverse Reinforcement Learning (ML-IRL) with Dynamically Truncated Soft Q-learning nested

Input:

Reward parameter θ_0; Explorative policy π_0; Stepsize α;

initialize $b_r \sim \text{Unif}(\{-1, 1\}), W_r(0) \sim N(0, I_d/d)\,(r \in [m]), \bar{W} = W(0), S_B = \left\{W \in \mathbb{R}^{md} : \|W - W(0)\|_2 \leq B\right\}\,(B > 0)$;

for $k = 0, 1, \ldots, K-1$ **do**

  Policy Evaluation $(k+2)$

  Policy Improvement: $\pi_{k+1}(\cdot \mid s) \propto \exp\left(Q^{soft}_{r_{\theta_k}, \pi_k}(s, \cdot)\right), \forall s \in \mathcal{S}$.

  Data Sampling I: Sampling expert trajectory $\tau_k^E := \{s_t, a_t\}_{t \geq 0}$ from the dataset D

  Data Sampling II: Sampling agent trajectory $\tau_k^A := \{s_t, a_t\}_{t \geq 0}$ from the policy π_{k+1}

  Estimating Gradient: $g_k := h\left(\theta_k; \tau_k^E\right) - h\left(\theta_k; \tau_k^A\right)$ where $h(\theta; \tau) := \sum_{t \geq 0} \gamma^t \nabla_\theta r\left(s_t, a_t; \theta\right)$

  Reward Parameter Update: $\theta_{k+1} := \theta_k + \alpha g_k$

end for

return π_k, θ_k;

3.3 Single Loop ML-IRL

As is discussed above, the policy evaluation step in Algorithm 2 still employs a nested inner loop. It therefore results in an overall complexity of $\mathcal{O}\left(K^2\right)$. To alleviate the computational burden, we design a novel purely single-loop Algorithm 3. Every round of policy improvement requires only one iteration of stochastic update in neural soft Q-learning to get the approximate soft Q value. Plus, the stepsize η_k is varying with iterations. Due to the fact that it only iterates once to approximately soft Q-value, the estimation could be very rough, especially in the beginning. Therefore, we decide to use a larger learning rate in the beginning training rounds. It is reasonable to do so because the policy estimation error is initially larger due to random initialization. To ensure global convergence, the choice of step size should be carefully selected. We hope to make inner policy evaluation converge faster than the reward update asymptotically. (For more details, please see Appendix 6.3.2) For detailed implementations of Algorithm 3, please check below.

4 Theoretical Analysis

In this section, we mainly provide theoretical insights into our proposed algorithms.

Algorithm 3 Single Loop Maximum Likelihood Inverse Reinforcement Learning (ML-IRL)

Input:

Reward parameter θ_0; Explorative policy π_0; Stepsize α; Critic decreasing step length sequence $\{\eta_k\}$.

initialize $b_r \sim \text{Unif}(\{-1,1\}), W_r(0) \sim N(0, I_d/d)\,(r \in [m]), \bar{W} = W(0), S_B = \left\{W \in \mathbb{R}^{md} : \|W - W(0)\|_2 \leq B\right\}\,(B > 0)$;

for $k = 0, 1, \ldots, K-1$ **do**

  Sample a tuple (s, a, r, s') from the stationary distribution μ_k of the exploration policy π_k

  Bellman residual calculation: $\delta \leftarrow \widehat{Q}(s, a; W(k)) - r_{\theta_k} - \gamma\, \text{softmax}_{a' \in \mathcal{A}}\, \widehat{Q}\left(s', a'; W(k)\right)$;

  TD update: $\widetilde{W}(k+1) \leftarrow W(k) - \eta_k \delta \cdot \nabla_W \widehat{Q}(s, a; W(k))$;

  Projection: $W(k+1) \leftarrow \text{argmin}_{W \in S_B} \|W - \widetilde{W}(k+1)\|_2$

  Averaging: $\bar{W} \leftarrow \frac{k+1}{k+2} \cdot \bar{W} + \frac{1}{k+2} \cdot W(k+1)$

  Output: $Q^{soft}_{r_{\theta_k}, \pi_k}(s, \cdot) = \widehat{Q}_{\text{out}}(\cdot) \leftarrow \widehat{Q}(\cdot; \bar{W})$

  Policy Improvement: $\pi_{k+1}(\cdot \mid s) \propto \exp\left(Q^{soft}_{r_{\theta_k}, \pi_k}(s, \cdot)\right), \forall s \in \mathcal{S}$.

  Data Sampling I: Sampling expert trajectory $\tau_k^E := \{s_t, a_t\}_{t \geq 0}$ from the dataset D

  Data Sampling II: Sampling agent trajectory $\tau_k^A := \{s_t, a_t\}_{t \geq 0}$ from the policy π_{k+1}

  Estimating Gradient: $g_k := h\left(\theta_k; \tau_k^E\right) - h\left(\theta_k; \tau_k^A\right)$ where $h(\theta; \tau) := \sum_{t \geq 0} \gamma^t \nabla_\theta r\left(s_t, a_t; \theta\right)$

  Reward Parameter Update: $\theta_{k+1} := \theta_k + \alpha g_k$

end for

return π_k, θ_k;

4.1 Definition:

Recall that

$$\begin{aligned}\widehat{Q}(x; W) &= \frac{1}{\sqrt{m}} \sum_{r=1}^{m} b_r \sigma\left(W_r^\top x\right) \\ &= \frac{1}{\sqrt{m}} \sum_{r=1}^{m} b_r \mathbb{1}\left\{W_r^\top x > 0\right\} W_r^\top x\end{aligned}$$

and $\nabla_{W_r} \widehat{Q}(x; W) = b_r \mathbb{1}\left\{W_r^\top x > 0\right\} x$ almost everywhere in $\mathbb{R}^{md}$.

Definition 1 *(Local Linearization of soft Q function)*

$$\widehat{Q}_0(x; W) = \Phi(x)^\top W \tag{6}$$

where $\Phi(x) = \frac{1}{\sqrt{m}} \cdot \left(\mathbb{1}\left\{W_1(0)^\top x > 0\right\} x, \ldots, \mathbb{1}\left\{W_m(0)^\top x > 0\right\} x\right) \in \mathbb{R}^{md}$,

which is linear in the feature map $\Phi(x)$. In other words, with respect to W, $\widehat{Q}_0(x; W)$ linearizes the neural network $\widehat{Q}(x; W)$ defined in (4) locally at random initialization $W(0)$.

Similarly, we have:

Definition 2 *(Local Linearization of soft Q function)*

$$r_0(x; \theta) = \Psi(x)^\top \theta \tag{7}$$

where $\Psi(x) = \frac{1}{\sqrt{m}} \cdot \left(\mathbb{1}\left\{\theta_1(0)^\top x > 0\right\} x, \ldots, \mathbb{1}\left\{\theta_m(0)^\top x > 0\right\} x\right) \in \mathbb{R}^{md}$.

4.2 Assumptions

Assumption 4.1 *(Ergodicity). For any policy π, assume the Markov chain with transition kernel $\mathcal{P}$ is irreducible and aperiodic under policy π. Then there exist constants $\kappa > 0$ and $\mu \in (0, 1)$ such*

that

$$\sup_{s\in\mathcal{S}} \|\mathbb{P}\left(s_t \in \cdot \mid s_0 = s, \pi\right) - \mu_\pi(\cdot)\|_{TV} \leq \kappa\mu^t, \quad \forall t \geq 0$$

where $\|\cdot\|_{TV}$ *is the total variation (TV) norm;* μ_π *is the stationary state distribution under* π*.*

Assumption 4.1 assumes the Markov chain mixes at a geometric rate. It is a common assumption in the iterature of RL Wu et al. (2022); Bhandari et al. (2018); Zou et al. (2019), which holds for any time-homogeneous Markov chain with finite-state space or any uniformly ergodic Markov chain with general state space.

Assumption 4.2 *(Regularity of Policy) There exists a constant* $\nu' > 0$ *such that for any* $W_1, W_2 \in S_B$*, it holds that*

$$\begin{aligned}&(\gamma+\nu')^{-2}\,\mathbb{E}_{x\sim\pi_k}\left[\left(\widehat{Q}_0\left(x;W_1\right)-\widehat{Q}_0\left(x;W_2\right)\right)^2\right]\\ &\geq \mathbb{E}_{s\sim\mu_k}\left[\left(\operatorname*{softmax}_{a\in\mathcal{A}}\widehat{Q}_0\left(s,a;W_1\right)-\operatorname*{softmax}_{a\in\mathcal{A}}\widehat{Q}_0\left(s,a;W_2\right)\right)^2\right]\end{aligned} \tag{8}$$

This assumption is proposed by Wang et al. (2019) for soft Q-learning convergence analysis. Zou et al. (2019); Melo et al. (2008)'s discussions on the global convergence of linear Q-learning relied on an assumption that implies 4.2.

Assumption 4.3 *(Regularity of Stationary Distribution* $\mu_{\exp}$*) There exists a constant* $c_1 > 0$ *such that for any* $\tau \geq 0$ *and* $w \in \mathbb{R}^d$ *with* $\|w\|_2 = 1$*, it holds that*

$$\mathbb{P}\left(\left|w^\top\psi(s,a)\right| \leq \tau, \textit{ for all } a \in \mathcal{A}\right) \leq c_1 \cdot \tau$$

where $(s,a) \sim \mu_k$*.*

The assumption of regularity (4.3) places a constraint on the density of μ concerning the marginal distribution of x. This constraint is naturally satisfied when the marginal distribution of x demonstrates a uniformly bounded probability density across the unit sphere, as discussed in Wang et al. (2019).

4.3 Lemmas

Lemma 1

$$L\left(\theta\right) = \mathbb{E}_{\tau\sim D}\left[\sum_{t=0}^{\infty}\gamma^t r\left(s_t,a_t;\theta\right)\right] - \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{\infty}\gamma^t\left(r\left(\mathbf{s}_t,\mathbf{a}_t;\theta\right)+\mathcal{H}\left(\pi\left(\cdot\mid\mathbf{s}_t\right)\right)\right)\right]. \tag{9}$$

Based on *soft policy iteration*, we could establish the following lemma to express the objective function in the above form for the convenience of analysis.

Lemma 2 *(Lemma 4.1 in Zeng et al. (2022b)) The gradient of the likelihood objective* $L(\theta)$ *in (2) can be equivalently expressed as follows:*

$$\nabla L(\theta) = \mathbb{E}_{\tau\sim\mathcal{D}}\left[\sum_{t\geq 0}\gamma^t\nabla_\theta r\left(s_t,a_t;\theta\right)\right] - \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t\geq 0}\gamma^t\nabla_\theta r\left(s_t,a_t;\theta\right)\right]. \tag{10}$$

To further investigate the theoretical properties of likelihood objective $L(\theta)$, we first introduce the following assumptions:

Lemma 3 *(Proposition 1 in Scaman and Virmaux (2019) and Lemma Gaur et al. (2023)) Let reward function be defined as (5) and soft Q-function be defined as (4). For any* $s \in \mathcal{S}, a \in \mathcal{A}$ *and any reward parameter* θ*, the following holds:*

$$\left|r(s,a;\theta_1) - r_(s,a;\theta_2)\right| \leq L_r\left\|\theta_1-\theta_2\right\| \tag{11}$$

$$\left|\hat{Q}(s,a;W_1) - \hat{Q}_(s,a;W_2)\right| \leq L_q\left\|W_1-W_2\right\| \tag{12}$$

where L_r, L_q *is a positive constant*

This lemma shows that the parameterized reward and soft Q-function have a bounded gradient, and are therefore Lipschitz continuous.

Lemma 4 *$V(\theta) = r(s_t, a_t; \theta) - r_0(s_t, a_t; \theta), r_0(s_t, a_t; \theta)$ is defined in (7)*

$$\nabla V(\theta) = \mathcal{O}\left(m^{-1/2}\right) \tag{13}$$

We establish this lemma to characterize the gradient error bound of local linearization of the reward function, which is critical to the next proposition.

Lemma 5

$$\|\nabla L(\theta_1) - \nabla L(\theta_2)\| \leq \mathcal{O}\left(m^{-1/2}\right) + L_c \|\theta_1 - \theta_2\| \tag{14}$$

The Lipschitz smoothness property is common in the literature of min-max / bi-level optimization Hong et al. (2020); Khanduri et al. (2021); Jin et al. (2020); Guan et al. (2021); Chen et al. (2021). We get an approximate Lipschitz smoothness property by studying the local linearization of the reward function.

Lemma 6 *(Xu et al. (2020),Lemma 3) Consider the initialization distribution $\eta(\cdot)$ and transition kernel $\mathcal{P}(\cdot \mid s, a)$. Under $\eta(\cdot)$ and $\mathcal{P}(\cdot \mid s, a)$, denote $d_w(\cdot, \cdot)$ as the state-action visitation distribution of MDP with the Boltzmann policy parameterized by parameter w. For all policy parameter w and w', we have*

$$\|d_w(\cdot, \cdot) - d_{w'}(\cdot, \cdot)\|_{TV} \leq C_d \|w - w'\|$$

where C_d is a positive constant.

Lemma 7

$$\mathbb{E}_{init,\mu}\left[|\delta(x, r, x'; W(k))|^2\right] \left\|\nabla_W \widehat{Q}(s, a; W(k))\right\|_2^2 \leq \mathcal{O}\left(B^2\right) \tag{15}$$

This Lemma establishes an upper bound of variance of stochastic updates in policy evaluation.

Lemma 8 *$g_k := h\left(\theta_k; \tau_k^E\right) - h\left(\theta_k; \tau_k^A\right)$ where $h(\theta; \tau) := \sum_{t \geq 0} \gamma^t \nabla_\theta r(s_t, a_t; \theta)$*

$$\|g_k\| \leq 2L_q \tag{16}$$

This lemma provides a constant upper Bound of stochastic updates of reward.

The following two lemmas are first established in Cai et al. (2023) on studying the convergence of neural soft Q-learning. We'll use them in our lower-level analysis.

Lemma 9 *Descent Lemma in soft Q-function (C.27 in Cai et al. (2023))*

$$\begin{aligned}
&\mathbb{E}_{init,,\mu}\left[\left(\widehat{Q}_0(x; W(k)) - \widehat{Q}_0(x; W^*)\right)^2\right] \\
&\leq \frac{\mathbb{E}_{init}\left[\|W(k) - W^*\|_2^2\right] - \mathbb{E}_{init}\left[\|W(k+1) - W^*\|_2^2\right] + \eta_k^2 \sigma_z^2}{(2\eta_k(1-\gamma) - 8\eta_k^2)} + O\left(B^3 m^{-1/2} + B^{5/2} m^{-1/4}\right)
\end{aligned} \tag{17}$$

Lemma 10 *Convergence of Stochastic Update (Theorem E.2 in Cai et al. (2023)) We set η to be of order $T^{-1/2}$ in Algorithm 1. Under Assumptions 4.2 and 4.3, the output $\widehat{Q}_{out}$ of Algorithm 1 satisfies*

$$\mathbb{E}_{init,\mu_{exp}}\left[\left(\widehat{Q}_{out}(x) - \widehat{Q}_0(x; W^*)\right)^2\right] = O\left(B^2 T^{-1/2} + B^3 m^{-1/2} + B^{5/2} m^{-1/4}\right). \tag{18}$$

T represents the number of iterations.

4.4 Theorem

Theorem 1 *Suppose Assumptions 4.1 hold. Selecting stepsize $\alpha := \frac{\alpha_0}{K^\sigma}$ for the reward update step and fixed $\eta_k = \min\{K^{-\frac{1}{2}}, (1-\gamma)/8\}$ for the policy evaluation update step where $\alpha_0 > 0$ and $\sigma \in (0,1)$ are some fixed constants, and K is the total number of iterations to be run by the Algorithm 2. Then the following result holds:*

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\left[\|\log\pi_{k+1} - \log\pi_{\theta_k}\|_\infty\right] = \mathcal{O}\left(K^{-\frac{1}{4}}\right) + O\left(B^3 m^{-1/2} + B^{5/2} m^{-1/4}\right) \tag{19}$$

$$\begin{aligned}\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\left[\|\nabla L(\theta_k)\|^2\right] &= \mathcal{O}\left(K^{-\sigma}\right) + \mathcal{O}\left(K^{-1+\sigma}\right) + \mathcal{O}\left(K^{-\frac{1}{4}}\right) \\ &\quad + O\left(m^{-1/2} + B^{3/2} m^{-1/4} + B^{5/4} m^{-1/8}\right)\end{aligned} \tag{20}$$

where we denote $\|\log\pi_{k+1} - \log\pi_{\theta_k}\|_\infty \quad := \max_{s\in\mathcal{S}, a\in\mathcal{A}} |\log\pi_{k+1}(a \mid s) - \log\pi_{\theta_k}(a \mid s)|$.

Theorem 2 *Suppose Assumptions 4.1 hold. Selecting stepsize $\alpha := \frac{\alpha_0}{K^\sigma}$ for the reward update step and $\eta_k = \min\{k^{-\frac{3}{4}}, (1-\gamma)/8\}$ for the policy evaluation update step where $\alpha_0 > 0$ and $\sigma \in (0,1)$ are some fixed constants, and K is the total number of iterations to be run by the Algorithm 2. Then the following result holds:*

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\left[\|\log\pi_{k+1} - \log\pi_{\theta_k}\|_\infty\right] = \mathcal{O}\left(K^{-\frac{1}{8}}\right) + O\left(B^3 m^{-1/2} + B^{5/2} m^{-1/4}\right) \tag{21}$$

$$\begin{aligned}\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\left[\|\nabla L(\theta_k)\|^2\right] &= \mathcal{O}\left(K^{-\sigma}\right) + \mathcal{O}\left(K^{-1+\sigma}\right) + \mathcal{O}\left(K^{-\frac{1}{8}}\right) \\ &\quad + O\left(m^{-1/2} + B^{3/2} m^{-1/4} + B^{5/4} m^{-1/8}\right)\end{aligned}$$

(22)

where we denote $\|\log\pi_{k+1} - \log\pi_{\theta_k}\|_\infty \quad := \max_{s\in\mathcal{S}, a\in\mathcal{A}} |\log\pi_{k+1}(a \mid s) - \log\pi_{\theta_k}(a \mid s)|$.

To study the global optimality of our algorithm, We translate the bi-level problem formulation to the following equivalent saddle point problem Zeng et al. (2023).

$$\begin{aligned}&\max_\theta \min_\pi L(\theta, \pi) := \\ &\mathbb{E}_{\tau^{\mathrm{E}}\sim(\eta,\pi^{\mathrm{E}},P)}\left[\sum_{t=0}^{\infty}\gamma^t\left(r\left(s_t, a_t; \theta\right)\right)\right] - \mathbb{E}_{\tau^{\mathrm{A}}\sim(\eta,\pi,\widehat{P})}\left[\sum_{t=0}^{\infty}\gamma^t\left(r\left(s_t, a_t; \theta\right) + \mathcal{H}\left(\pi\left(\cdot \mid s_t\right)\right)\right)\right].\end{aligned} \tag{23}$$

Theorem 3 *(Approximate Concavity) For any $\theta' \in S_{B_\theta}$ that*

$$\mathbb{E}_{init}\left[L\left(\pi_{\theta_k}, \theta'\right) - L\left(\pi_{\theta_k}, \theta_k\right)\right] = \mathbb{E}_{init}\left[\nabla_\theta L\left(\pi_{\theta_k}, \theta_k\right)^\top\left(\theta'_k - \theta_k\right)\right] + \mathcal{O}\left(B^{3/2}\cdot m^{-1/4}\right) \tag{24}$$

Therefore, L is concave in the reward parameter under the over-parametrized case, so we have

$$\tilde{\theta} \in \arg\max_\theta \min_{\pi_\theta} L(\theta, \pi) = \arg\max_\theta L(\theta). \tag{25}$$

we obtain that for any stationary point of the objective $L(\cdot)$ and its corresponding policy in problem (3) constructs a global optimal point $\left(\tilde{\theta}, \pi_{\tilde{\theta}}\right)$ of $L(\cdot,\cdot)$.

5 Conclusions

In this paper, we improve the maximum likelihood IRL framework and propose two provably efficient algorithms with finite-time guarantees. To our knowledge, Algorithm 3 is the first authentic single-loop algorithm to theoretically address the neural IRL problem. Moreover, we provide the first non-asymptotic analysis for the IRL algorithm under neural network parameterization. Algorithm 2 and 3 can both efficiently find the global optimum with finite-time guarantees. A limitation of our method is the requirement of a sufficiently large width of the parametrization neural network, so one future direction of this work is to further extend our analysis to finite-width case.

References

Allen-Zhu, Z., Li, Y., and Liang, Y. (2020). Learning and generalization in overparameterized neural networks, going beyond two layers.

Arora, S., Du, S. S., Hu, W., Li, Z., and Wang, R. (2019). Fine-grained analysis of optimization and generalization for overparameterized two-layer neural networks.

Bhandari, J., Russo, D., and Singal, R. (2018). A finite time analysis of temporal difference learning with linear function approximation.

Bloem, M. and Bambos, N. (2014). Infinite time horizon maximum causal entropy inverse reinforcement learning. In *53rd IEEE conference on decision and control*, pages 4911–4916. IEEE.

Borkar, V. S. (1997). Stochastic approximation with two time scales. *Systems & Control Letters*, 29(5):291–294.

Cai, Q., Yang, Z., Lee, J. D., and Wang, Z. (2023). Neural temporal difference and q learning provably converge to global optima. *Mathematics of Operations Research*.

Cen, S., Cheng, C., Chen, Y., Wei, Y., and Chi, Y. (2022). Fast global convergence of natural policy gradient methods with entropy regularization. *Operations Research*, 70(4):2563–2578.

Chen, T., Sun, Y., and Yin, W. (2021). Closing the gap: Tighter analysis of alternating stochastic gradient methods for bilevel problems. *Advances in Neural Information Processing Systems*, 34:25294–25307.

Du, S. and Lee, J. (2018). On the power of over-parametrization in neural networks with quadratic activation. In *International conference on machine learning*, pages 1329–1338. PMLR.

Du, S. S., Zhai, X., Poczos, B., and Singh, A. (2018). Gradient descent provably optimizes over-parameterized neural networks. *arXiv preprint arXiv:1810.02054*.

Fu, J., Korattikara, A., Levine, S., and Guadarrama, S. (2019). From language to goals: Inverse reinforcement learning for vision-based instruction following. In *International Conference on Learning Representations*.

Fu, Z., Yang, Z., and Wang, Z. (2021). Single-timescale actor-critic provably finds globally optimal policy.

Gaur, M., Bedi, A. S., Wang, D., and Aggarwal, V. (2023). On the global convergence of natural actor-critic with two-layer neural network parametrization. *arXiv preprint arXiv:2306.10486*.

Guan, Z., Xu, T., and Liang, Y. (2021). When will generative adversarial imitation learning algorithms attain global convergence. In *International Conference on Artificial Intelligence and Statistics*, pages 1117–1125. PMLR.

Haarnoja, T., Tang, H., Abbeel, P., and Levine, S. (2017). Reinforcement learning with deep energy-based policies. In *International conference on machine learning*, pages 1352–1361. PMLR.

Haarnoja, T., Zhou, A., Abbeel, P., and Levine, S. (2018). Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International conference on machine learning*, pages 1861–1870. PMLR.

Hanin, B. (2019). Universal function approximation by deep neural nets with bounded width and relu activations. *Mathematics*, 7(10):992.

Hanin, B. and Sellke, M. (2017). Approximating continuous functions by relu nets of minimal width. *arXiv preprint arXiv:1710.11278*.

Hofmann, T., Schölkopf, B., and Smola, A. J. (2008). Kernel methods in machine learning.

Hong, M., Wai, H., Wang, Z., and Yang, Z. (2020). A two-timescale framework for bilevel optimization: Complexity analysis and application to actor-critic. arxiv e-prints, art. *arXiv preprint arXiv:2007.05170*.

Ji, K., Yang, J., and Liang, Y. (2021). Bilevel optimization: Convergence analysis and enhanced design. In *International conference on machine learning*, pages 4882–4892. PMLR.

Jin, C., Netrapalli, P., and Jordan, M. (2020). What is local optimality in nonconvex-nonconcave minimax optimization? In *International conference on machine learning*, pages 4880–4889. PMLR.

Khanduri, P., Zeng, S., Hong, M., Wai, H.-T., Wang, Z., and Yang, Z. (2021). A near-optimal algorithm for stochastic bilevel optimization via double-momentum. *Advances in neural information processing systems*, 34:30271–30283.

Kim, K., Garg, S., Shiragur, K., and Ermon, S. (2021). Reward identification in inverse reinforcement learning. In *International Conference on Machine Learning*, pages 5496–5505. PMLR.

Li, Y. and Liang, Y. (2018). Learning overparameterized neural networks via stochastic gradient descent on structured data. *Advances in neural information processing systems*, 31.

Liang, S. and Srikant, R. (2016). Why deep neural networks for function approximation? *arXiv preprint arXiv:1610.04161*.

Lillicrap, T. P., Hunt, J. J., Pritzel, A., Heess, N., Erez, T., Tassa, Y., Silver, D., and Wierstra, D. (2015). Continuous control with deep reinforcement learning. *arXiv preprint arXiv:1509.02971*.

Lu, Z., Pu, H., Wang, F., Hu, Z., and Wang, L. (2017). The expressive power of neural networks: A view from the width. *Advances in neural information processing systems*, 30.

Maei, H., Szepesvari, C., Bhatnagar, S., Precup, D., Silver, D., and Sutton, R. S. (2009). Convergent temporal-difference learning with arbitrary smooth function approximation. *Advances in neural information processing systems*, 22.

Melo, F. S., Meyn, S. P., and Ribeiro, M. I. (2008). An analysis of reinforcement learning with function approximation. In *Proceedings of the 25th international conference on Machine learning*, pages 664–671.

Mnih, V., Badia, A. P., Mirza, M., Graves, A., Lillicrap, T., Harley, T., Silver, D., and Kavukcuoglu, K. (2016). Asynchronous methods for deep reinforcement learning. In *International conference on machine learning*, pages 1928–1937. PMLR.

Osa, T., Pajarinen, J., Neumann, G., Bagnell, J. A., Abbeel, P., and Peters, J. (2018). An algorithmic perspective on imitation learning. *Foundations and Trends in Robotics*, 7(1-2):1–179.

Scaman, K. and Virmaux, A. (2019). Lipschitz regularity of deep neural networks: analysis and efficient estimation.

Sutton, R. S. (1988). Learning to predict by the methods of temporal differences. *Machine learning*, 3:9–44.

Telgarsky, M. (2015). Representation benefits of deep feedforward networks. *arXiv preprint arXiv:1509.08101*.

Telgarsky, M. (2016). Benefits of depth in neural networks. In *Conference on learning theory*, pages 1517–1539. PMLR.

Wang, L., Cai, Q., Yang, Z., and Wang, Z. (2019). Neural policy gradient methods: Global optimality and rates of convergence. *arXiv preprint arXiv:1909.01150*.

Wu, Y., Zhang, W., Xu, P., and Gu, Q. (2022). A finite time analysis of two time-scale actor critic methods.

Wu, Z., Sun, L., Zhan, W., Yang, C., and Tomizuka, M. (2020). Efficient sampling-based maximum entropy inverse reinforcement learning with application to autonomous driving.

Wulfmeier, M., Ondruska, P., and Posner, I. (2015). Maximum entropy deep inverse reinforcement learning. *arXiv preprint arXiv:1507.04888*.

Xu, T., Wang, Z., and Liang, Y. (2020). Improving sample complexity bounds for (natural) actor-critic algorithms. *Advances in Neural Information Processing Systems*, 33:4358–4369.

Yarotsky, D. (2017). Error bounds for approximations with deep relu networks. *Neural Networks*, 94:103–114.

Yarotsky, D. (2018). Optimal approximation of continuous functions by very deep relu networks. In *Conference on learning theory*, pages 639–649. PMLR.

Zeng, S., Hong, M., and Garcia, A. (2022a). Structural estimation of markov decision processes in high-dimensional state space with finite-time guarantees. *arXiv preprint arXiv:2210.01282*.

Zeng, S., Li, C., Garcia, A., and Hong, M. (2022b). Maximum-likelihood inverse reinforcement learning with finite-time guarantees. *Advances in Neural Information Processing Systems*, 35:10122–10135.

Zeng, S., Li, C., Garcia, A., and Hong, M. (2023). Understanding expertise through demonstrations: A maximum likelihood framework for offline inverse reinforcement learning. *arXiv preprint arXiv:2302.07457*.

Zhou, L. and Small, K. (2020). Inverse reinforcement learning with natural language goals.

Ziebart, B. D., Bagnell, J. A., and Dey, A. K. (2010). Modeling interaction via the principle of maximum causal entropy. In *International Conference on Machine Learning*.

Ziebart, B. D., Bagnell, J. A., and Dey, A. K. (2013). The principle of maximum causal entropy for estimating interacting processes. *IEEE Transactions on Information Theory*, 59(4):1966–1980.

Ziebart, B. D., Maas, A. L., Bagnell, J. A., Dey, A. K., et al. (2008). Maximum entropy inverse reinforcement learning. In *Aaai*, volume 8, pages 1433–1438. Chicago, IL, USA.

Zou, D., Cao, Y., Zhou, D., and Gu, Q. (2018). Stochastic gradient descent optimizes over-parameterized deep relu networks. arxiv e-prints, art. *arXiv preprint arXiv:1811.08888*.

Zou, S., Xu, T., and Liang, Y. (2019). Finite-sample analysis for sarsa with linear function approximation.

6 Supplementary Material

6.1 Proof of Lemmas

6.1.1 Proof of Lemma 1

First, we are able to express the objective function $L(\theta)$ in (2) as below:

$$L(\theta) := \mathbb{E}_{\tau\sim D}\left[\sum_{t=0}^{\infty}\gamma^t \log \pi_\theta\left(a_t \mid s_t\right)\right] \overset{(i)}{=} \mathbb{E}_{\tau\sim D}\left[\sum_{k=0}^{\infty}\gamma^t \log\left(\frac{\exp\left(Q^{\text{soft}}_{r_\theta,\pi_\theta}\left(s_t,a_t\right)\right)}{\sum_a \exp\left(Q^{\text{soft}}_{r_\theta,\pi_\theta}\left(s_t,a\right)\right)}\right)\right]$$

where (i) is due to the fact that the optimal policy has the closed form $\pi_\theta(\cdot \mid s) \propto \exp\left(Q^{\text{sot}}_{r_\theta,\pi_\theta}(s,\cdot)\right)$. Therefore, we could express the objective function in this form:

$$
\begin{aligned}
L(\theta) &:= \mathbb{E}_{\tau\sim D}\left[\sum_{t=0}^{\infty}\gamma^t\left(Q^{\text{soft}}_{r_\theta,\pi_\theta}\left(s_t,a_t\right) - \log\left(\sum_a \exp\left(Q^{\text{soft}}_{r_\theta,\pi_\theta}\left(s_t,a\right)\right)\right)\right)\right] \\
&\overset{(i)}{=} \mathbb{E}_{\tau\sim D}\left[\sum_{t=0}^{\infty}\gamma^t\left(Q^{\text{soft}}_{r_\theta,\pi_\theta}\left(s_t,a_t\right) - V^{\text{soft}}_{r_\theta,\pi_\theta}\left(s_t\right)\right)\right] \\
&= \mathbb{E}_{\tau\sim D}\left[\sum_{t=0}^{\infty}\gamma^t\left(r\left(s_t,a_t;\theta\right) + \gamma V^{\text{soft}}_{r_\theta,\pi_\theta}\left(s_{t+1}\right) - V^{\text{soft}}_{r_\theta,\pi_\theta}\left(s_t\right)\right)\right] \\
&= \mathbb{E}_{\tau\sim D}\left[\sum_{t=0}^{\infty}\gamma^t r\left(s_t,a_t;\theta\right)\right] + \mathbb{E}_{\tau\sim D}\left[\sum_{t=1}^{\infty}\gamma^t V^{\text{soft}}_{r_\theta,\pi_\theta}\left(s_t\right)\right] - \mathbb{E}_{\tau\sim D}\left[\sum_{t=0}^{\infty}\gamma^t V^{\text{soft}}_{r_\theta,\pi_\theta}\left(s_t\right)\right] \\
&= \mathbb{E}_{\tau\sim D}\left[\sum_{t=0}^{\infty}\gamma^t r\left(s_t,a_t;\theta\right)\right] - \mathbb{E}_{s_0\sim\eta(\cdot)}\left[V^{\text{soft}}_{r_\theta,\pi_\theta}\left(s_0\right)\right] \\
&= \mathbb{E}_{\tau\sim D}\left[\sum_{t=0}^{\infty}\gamma^t r\left(s_t,a_t;\theta\right)\right] - \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{\infty}\gamma^t\left(r\left(\mathbf{s}_t,\mathbf{a}_t;\theta\right) + \mathcal{H}\left(\pi\left(\cdot \mid \mathbf{s}_t\right)\right)\right)\right]
\end{aligned}
$$

where (i) follows the fact that the optimal soft value function could be expressed as $V^{\text{soft}}_{r_\theta,\pi_\theta}(s) = \log\left(\sum_a \exp\left(Q^{\text{soft}}_{r_\theta,\pi_\theta}(s,a)\right)\right)$.

6.1.2 Proof of Lemma 4

First, we prove $V(\theta)$ is Lipschitz continuous.

$$
\begin{aligned}
\left\|V(\theta) - h(\theta')\right\|_2 &\leq \left\|r(s_t,a_t;\theta) - r(s_t,a_t;\theta')\right\|_2 + \left\|r_0(s_t,a_t;\theta) - r_0(s_t,a_t;\theta')\right\|_2 \\
&\overset{(\text{i})}{\leq} \frac{2}{\sqrt{m}}\left\|\theta - \theta'\right\|_2
\end{aligned} \tag{26}
$$

where (i) is using the property from equation (11) in Lemma 3.
Now we need to show $\nabla V(\theta)$ is bounded:

$$
\left|\nabla V(\theta)\right| = \lim_{\delta\to 0}\frac{\left|h(\theta+\delta) - V(\theta)\right|}{|\delta|} \leq \lim_{\delta\to 0}\frac{\frac{2}{\sqrt{m}}\left|\theta+\delta-\theta\right|}{|\delta|} = \frac{2}{\sqrt{m}} \tag{27}
$$

Therefore, the lemma is shown.

6.1.3 Proof of Lemma 5

First we examine:

$$
\begin{aligned}
&\nabla_\theta\left[\mathbb{E}_{\tau\sim\mathcal{D}}\left[r\left(s_t,a_t;\theta\right)\right] - \mathbb{E}_{\tau\sim\pi_\theta}\left[r\left(s_t,a_t;\theta\right)\right]\right] \\
&= -\underbrace{\nabla_\theta\left[\mathbb{E}_{\tau\sim\pi_\theta}\left[r\left(s_t,a_t;\theta\right)\right] - \mathbb{E}_{\tau\sim\pi_\theta}\left[r_0\left(s_t,a_t;\theta\right)\right]\right]}_{\text{(i)}} - \underbrace{\nabla_\theta\left[\mathbb{E}_{\tau\sim\pi_\theta}\left[r_0\left(s_t,a_t;\theta\right)\right] - \mathbb{E}_{\tau\sim\mathcal{D}}\left[r_0\left(s_t,a_t;\theta\right)\right]\right]}_{\text{(ii)}} \\
&\quad + \underbrace{\nabla_\theta\left[\mathbb{E}_{\tau\sim\mathcal{D}}\left[r\left(s_t,a_t;\theta\right)\right] - \mathbb{E}_{\tau\sim\mathcal{D}}\left[r_0\left(s_t,a_t;\theta\right)\right]\right]}_{\text{(iii)}} \\
&\overset{(\text{a})}{=} \nabla_\theta\left[\mathbb{E}_{\tau\sim\mathcal{D}}\left[r_0\left(s_t,a_t;\theta\right)\right] - \mathbb{E}_{\tau\sim\pi_\theta}\left[r_0\left(s_t,a_t;\theta\right)\right]\right] + \mathcal{O}\left(m^{-1/2}\right)
\end{aligned} \tag{28}
$$

where the equality (a) comes from Lemma 4. Term (i) and (iii) are globally bounded by $\mathcal{O}\left(m^{-1/2}\right)$.
Then we establish the following:

$$
\begin{aligned}
&\left\{\nabla_\theta\left[\mathbb{E}_{\tau\sim\mathcal{D}}\left[r\left(s_t,a_t;\theta\right)\right]-\mathbb{E}_{\tau\sim\pi_\theta}\left[r\left(s_t,a_t;\theta\right)\right]\right]\right\}-\left\{\nabla_\theta\left[\mathbb{E}_{\tau\sim\mathcal{D}}\left[r\left(s_t,a_t;\theta'\right)\right]-\mathbb{E}_{\tau\sim\pi_{\theta'}}\left[r\left(s_t,a_t;\theta'\right)\right]\right]\right\}\\
&\overset{(1)}{=}\nabla_\theta\left[\mathbb{E}_{\tau\sim\mathcal{D}}\left[r_0\left(s_t,a_t;\theta\right)\right]-\mathbb{E}_{\tau\sim\pi_\theta}\left[r_0\left(s_t,a_t;\theta\right)\right]\right]+\nabla_\theta\left[\mathbb{E}_{\tau\sim\pi'_\theta}\left[r_0\left(s_t,a_t;\theta\right)\right]-\mathbb{E}_{\tau\sim\mathcal{D}}\left[r_0\left(s_t,a_t;\theta'\right)\right]\right]\\
&+\mathcal{O}\left(m^{-1/2}\right)\\
&\overset{(2)}{=}\nabla_\theta\left\{\mathbb{E}_{\tau\sim\pi'_\theta}\left[r_0\left(s_t,a_t;\theta'\right)\right]-\mathbb{E}_{\tau\sim\pi_\theta}\left[r_0\left(s_t,a_t;\theta\right)\right]\right\}+\mathcal{O}\left(m^{-1/2}\right)\\
&\overset{(3)}{\leq}\max_{s\in\mathcal{S},a\in\mathcal{A}}\left\|\nabla_\theta r_0\left(s,a;\theta_k\right)\right\|\cdot\mathbb{E}\left[\left\|d\left(\cdot,\cdot;\pi_\theta\right)-d\left(\cdot,\cdot;\pi_{\theta'}\right)\right\|_{TV}\right]+\mathcal{O}\left(m^{-1/2}\right)\\
&\overset{(4)}{\leq}L_r\left\|Q^{\text{soft}}_{r_\theta,\pi_\theta}-Q^{\text{soft}}_{r_{\theta'},\pi_{\theta'}}\right\|+\mathcal{O}\left(m^{-1/2}\right)\\
&\overset{(5)}{\leq}L_rL_q\left\|\theta_1-\theta_2\right\|+\mathcal{O}\left(m^{-1/2}\right)
\end{aligned}
\tag{29}
$$

where (i) is from (28). We get the inequality (3) due to distribution mismatch between two visitation measures; the inequality (4) is derived through Lemma 3 and 6. The final (5) is the result of equation (12) in 3.

By Lemma 2, we have:
$\nabla L(\theta)=\mathbb{E}_{\tau\sim\mathcal{D}}\left[\sum_{t\geq0}\gamma^t\nabla_\theta r\left(s_t,a_t;\theta\right)\right]-\mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t\geq0}\gamma^t\nabla_\theta r\left(s_t,a_t;\theta\right)\right].$

By calculating the discounted sum of Equation (29), we obtain the approximate L-smoothness property:

$$
\left\|\nabla L\left(\theta_1\right)-\nabla L\left(\theta_2\right)\right\|\leq\mathcal{O}\left(m^{-1/2}\right)+L_c\left\|\theta_1-\theta_2\right\| \tag{30}
$$

where $L_c=\frac{1}{1-\gamma}L_rL_q$

6.1.4 Proof of Lemma 8

$\|g_k\|\leq\left\|h\left(\theta_k,\tau_k^E\right)\right\|+\left\|h\left(\theta_k,\tau_k^A\right)\right\|\leq2L_r\sum_{t\geq0}\gamma^t=\frac{2L_r}{1-\gamma}=2L_q$

6.2 Neural Soft Q-learning for Policy Evaluation

Definition 3 *The soft Bellman optimality operator is defined as:*

$$
\begin{aligned}
&\mathcal{T}_\beta(Q^{\text{soft}})(s,a)=\mathbb{E}\left[r(s,a)+\gamma\cdot\underset{a'\in\mathcal{A}}{\text{softmax}}Q^{\text{soft}}\left(s',a'\right)\mid s'\sim\mathcal{P}(\cdot\mid s,a)\right],\\
&\textit{where}\ \underset{a'\in\mathcal{A}}{\text{softmax}}Q^{\text{soft}}\left(s',a'\right)=\beta^{-1}\cdot\log\sum_{a'\in\mathcal{A}}\exp\left(\beta\cdot Q^{\text{soft}}\left(s',a'\right)\right),
\end{aligned}
\tag{31}
$$

for which Q^{soft}_π is the fixed point, that is,

$$
Q^{\text{soft}}=\mathcal{T}_\beta(Q^{\text{soft}}).
$$

To find the best function approximation of Q^{soft}_π, we usually turn to minimize a surrogate measure for the Mean Squared Bellman Error (MSBE), which is referred to as the Projected Mean-Squared Bellman Error (MSPBE). Wang et al. (2019)

$$
\min_W\text{MSPBE}(W)=\mathbb{E}_{(s,a)\sim\mu}\left[\left(\widehat{Q}(s,a;W)-\Pi_{\mathcal{F}}\mathcal{T}^\pi\widehat{Q}_\theta(s,a;W)\right)^2\right] \tag{32}
$$

where the soft Q-function is parametrized by $\widehat{Q}(s,a;W)$ with parameter W. In this context, μ signifies the stationary distribution of (s,a) corresponding to the policy π. The projection onto a function class $\mathcal{F}$ is represented as $\Pi_{\mathcal{F}}$.

For the nonlinear approximation of the Q-function, such as a neural network Maei et al. (2009) is utilized, the function class $\mathcal{F}$ is defined as $\left\{\widehat{Q}_\theta + \nabla_\theta \widehat{Q}_\theta^\top (\theta' - \theta) : \theta' \in \Theta\right\}$. $S_B = \left\{W \in \mathbb{R}^{md} : \|W - W(0)\|_2 \leq B\right\}$ is the parameter space of feasible W. The bounded domain limits the distance between the parameters W and the initial parameters $W(0)$, enhancing training stability.This class encompasses the local linearization of $\widehat{Q}_{W'}$ at the specific point W. We then define the function class

$$\mathcal{F}_{B,m}^\dagger = \left\{\frac{1}{\sqrt{m}} \sum_{r=1}^m b_r \mathbb{1}\left\{\left(W_r^\dagger\right)^\top x > 0\right\} W_r^\top x : W \in S_B\right\} \tag{33}$$

which consists of the local linearization of $\widehat{Q}(x; W)$ at $W = W^\dagger$.

Definition 4 *(Stationary Point of MSPBE) The stationary point $W^\dagger$ satisfies the following stationarity condition,*

$$\mathbb{E}_\mu\left[\delta\left(x, r, x'; W^\dagger\right) \cdot \nabla_W \widehat{Q}\left(x; W^\dagger\right)\right]^\top \left(W - W^\dagger\right) \geq 0, \tag{34}$$

for any $W \in S_B$. where we define

$$\delta_0\left(x, r, x'; W\right) = \widehat{Q}_0(x; W) - r - \gamma \widehat{Q}_0\left(x'; W\right)$$

Then (34) takes the following equivalent form

$$\left\langle \widehat{Q}\left(\cdot; W^\dagger\right) - \mathcal{T}^\pi \widehat{Q}\left(\cdot; W^\dagger\right), f(\cdot) - \widehat{Q}\left(\cdot; W^\dagger\right)\right\rangle_\mu \geq 0 \tag{35}$$

for any $f \in \mathcal{F}_{B,m}^\dagger$, which implies $\widehat{Q}\left(\cdot; W^\dagger\right) = \Pi_{\mathcal{F}_{B,m}^\dagger} \mathcal{T}_\beta \widehat{Q}\left(\cdot; W^\dagger\right)$ by the definition of the projection induced by $\langle \cdot, \cdot \rangle_\mu$. By (32), $\widehat{Q}\left(\cdot; W^\dagger\right)$ is the global optimum of the MSPBE that corresponds to the projection onto $\mathcal{F}_{B,m}^\dagger$.

Definition 5 *(Approximate Function Class)*

$$\mathcal{F}_{B,m} = \left\{\frac{1}{\sqrt{m}} \sum_{r=1}^m b_r \mathbb{1}\left\{W_r(0)^\top x > 0\right\} W_r^\top x : W \in S_B\right\}$$

By Cai et al. (2023), to analyze the global convergence of soft Q-learning when m is sufficiently large, we can replace $\mathcal{F}_{B,m}$ with $\mathcal{F}_{B,m}^\dagger$.This simplifies the analysis because we have knowledge of the distribution of $W(0)$. In pursuit of this objective, we introduce an approximate stationary point denoted as W^* concerning the function class $\mathcal{F}_{B,m}$, which is defined in Definition 5.

Lemma 11 *(Lemma 4.2 in Cai et al. (2023)) (Existence, Uniqueness, and Optimality of $\widehat{Q}_0\left(\cdot; W^*\right)$). There exists an approximate stationary point W^* for any $b \in \mathbb{R}^m$ and $W(0) \in \mathbb{R}^{md}$. Also, $\widehat{Q}_0\left(\cdot; W^*\right)$ is unique almost everywhere and is the global optimum of the MSPBE that corresponds to the projection onto $\mathcal{F}_{B,m}$.*

This lemma establishes the unique existence of an approximate stationary point, as it corresponds to the fixed point of the operator $\Pi_{\mathcal{F}_{B,m}} \mathcal{T}^\pi$. This projection operator is an ℓ_2-norm contraction, associated with the stationary distribution μ.

Lemma 12 *(Proposition 4.7 in Cai et al. (2023))*

$$\left\|\widehat{Q}_0\left(\cdot; W^*\right) - Q^\pi(\cdot)\right\|_\mu \leq (1-\gamma)^{-1} \cdot \left\|\Pi_{\mathcal{F}_{B,m}} Q^\pi(\cdot) - Q^\pi(\cdot)\right\|_\mu \tag{36}$$

When using an overparametrized neural network with width $m \to \infty$ and sufficiently large B, the projection $\Pi_{\mathcal{F}_{B,\infty}}$ reduces to identity. Hofmann et al. (2008); Cai et al. (2023). By Lemma 12, $\widehat{Q}_0\left(\cdot; W^*\right) = Q^\pi(\cdot)$, and $\widehat{Q}_0\left(\cdot; W^*\right)$ is equivalently the global optimum of the MSBE.

$$\mathbb{E}_\mu\left[\left(Q(x) - \mathcal{T}^\pi Q(x)\right)^2\right]$$

Therefore, it suffices to consider $\mathcal{F}_{B,m}$ in place of $\mathcal{F}_{B,m}^\dagger$ for our convergence analysis.

6.3 Lower-level Analysis

For any $s \in \mathcal{S}$ and $a \in \mathcal{A}$, we have the following relation:

$$\begin{aligned}
&|\log(\pi_{k+1}(a \mid s)) - \log(\pi_{\theta_k}(a \mid s))| \\
&\overset{(i)}{=} \left| \log\left(\frac{\exp\left(Q^{\text{soft}}_{r_{\theta_k},\pi_k}(s,a)\right)}{\sum_{\tilde{a}} \exp\left(Q^{\text{soft}}_{r_{\theta_k},\pi_k}(s,\tilde{a})\right)} \right) - \log\left(\frac{\exp\left(Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}(s,a)\right)}{\sum_{\tilde{a}} \exp\left(Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}(s,\tilde{a})\right)} \right) \right| \\
&\overset{(ii)}{\leq} \left| \log\left(\sum_{\tilde{a}} \exp\left(Q^{\text{soft}}_{r_{\theta_k},\pi_k}(s,\tilde{a})\right) \right) - \log\left(\sum_{\tilde{a}} \exp\left(Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}(s,\tilde{a})\right) \right) \right| \\
&+ \left| Q^{\text{soft}}_{r_{\theta_k},\pi_k}(s,a) - Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}(s,a) \right|
\end{aligned} \tag{37}$$

where (i) the fact that $\pi(a \mid s) \propto \exp\left(Q^{\text{soft}}(s,a)\right)$; (ii) follows the triangle inequality. We further analyze the second term in (37).

We first denote the operator $\log(\|\exp(v)\|_1) := \log\left(\left\|\sum_{\tilde{a}\in\mathcal{A}} \exp(v_{\tilde{a}})\right\|_1\right)$, where the vector $v \in \mathbb{R}^{|\mathcal{A}|}$ and $v = \left[v_1, v_2, \cdots, v_{|\mathcal{A}|}\right]$. Then for any $v', v'' \in \mathbb{R}^{|\mathcal{A}|}$, we have the following relation:

$$\begin{aligned}
|\log(\|\exp(v')\|_1) - \log(\|\exp(v'')\|_1) &\overset{(i)}{=} \langle v' - v'', \nabla_v \log(\|\exp(v)\|_1)|_{v=v^c} \rangle \\
&\leq \|v' - v''\|_\infty \cdot \|\nabla_v \log(\|\exp(v)\|_1)|_{v=v^c}\|_1 \\
&\overset{(ii)}{=} \|v' - v''\|_\infty
\end{aligned} \tag{38}$$

where (i) follows the mean value theorem and v_c is a convex combination of v' and v''; (ii) follows the following equalities:

$$[\nabla_v \log(\|\exp(v)\|_1)]_i = \frac{\exp(v_i)}{\sum_{1\leq a\leq|\mathcal{A}|} \exp(v_a)}, \quad \|\nabla_v \log(\|\exp(v)\|_1)\|_1 = 1, \quad \forall v \in \mathbb{R}^{|\mathcal{A}|}.$$

Through plugging (38)into (37), it holds that

$$\begin{aligned}
&|\log(\pi_{k+1}(a \mid s)) - \log(\pi_{\theta_k}(a \mid s))| \\
&\leq \left| Q^{\text{soft}}_{r_{\theta_k},\pi_k}(s,a) - Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}(s,a) \right| + \max_{\tilde{a}\in\mathcal{A}} \left| Q^{\text{soft}}_{r_{\theta_k},\pi_k}(s,\tilde{a}) - Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}(s,\tilde{a}) \right|
\end{aligned} \tag{39}$$

Taking the infinity norm over $\mathbb{R}^{|\mathcal{S}|\cdot|\mathcal{A}|}$, the following result holds:

$$\|\log \pi_{k+1} - \log \pi_{\theta_k}\|_\infty \leq 2 \left\| Q^{\text{soft}}_{r_{\theta_k},\pi_k} - Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}} \right\|_\infty \tag{40}$$

where $\|\log \pi_{k+1} - \log \pi_{\theta_k}\|_\infty = \max_{s\in\mathcal{S},a\in\mathcal{A}} |\log \pi_{k+1}(a \mid s) - \log \pi_{\theta_k}(a \mid s)|$ and $\left\| Q^{\text{soft}}_{r_{\theta_k},\pi_k} - Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}} \right\|_\infty = \max_{s\in\mathcal{S},a\in\mathcal{A}} \left| Q^{\text{soft}}_{r_{\theta_k},\pi_k}(s,a) - Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}(s,a) \right|$

Based on the inequality (40), we analyze $\left\| Q^{\text{soft}}_{r_{\theta_k},\pi_k} - Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}} \right\|_\infty$ to show the convergence of the policy parameter. It leads to the following analysis:

$$\left\| Q^{\text{soft}}_{r_{\theta_k},\pi_k} - Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}} \right\|_\infty \leq \left\| Q^{\text{soft}}_{r_{\theta_k},\pi_k} - Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}} \right\|_2 \tag{41}$$

Theorem 4 *(The Projection Theorem for Convex Sets). Let $y \in \mathbb{R}^n$ and let $C \subset \mathbb{R}^n$ be nonempty, closed and convex set. Then $\bar{x} \in C$ solves the problem*

$$\min\left\{ \frac{1}{2}\|y - x\|_2^2 : x \in C \right\}$$

if and only if $(y - \bar{x})^T(x - \bar{x}) \leq 0, \forall x \in C$. Moreover, the solution $\bar{x}$ always exists and is unique.

Proposition 1 $\Pi_{S_B}\left(W^* - \eta^* \cdot \bar{z}_0^*\right)$=$\Pi_{S_B}\left(W^* - \eta_k \cdot \bar{z}_0^*\right)$, $\forall k$

Proof 6.1 *From the left-hand side,* $\Pi_{S_B}\left(W^* - \eta^* \cdot \bar{z}_0^*\right)$=$W^*$ *by optimality. By Theorem 4, we have*

$$-\eta^* \cdot \bar{z}_0^*(W - W^*) \leq 0 \tag{42}$$

Since the step length η *is nonnegative, we have* $-\eta_k \cdot \bar{z}_0^*(W - W^*) \leq 0$*. Then we get the proposition.*

Lemma 13 *Stochastic Descent Lemma*

$$\begin{aligned}
&\mathbb{E}_{W,\mu}\left[\|W(t+1) - W^*\|_2^2\right] \\
&\leq \mathbb{E}_W\left[\|W(t) - W^*\|_2^2\right] - \left(2\eta_k(1-\gamma) - 8\eta_k^2\right) \cdot \mathbb{E}_{W,\mu}\left[\left(\widehat{Q}_0(x; W(k)) - \widehat{Q}_0\left(x; W^*\right)\right)^2\right] \\
&\quad + \underbrace{\mathbb{E}_W\left[2\eta_k^2 \cdot \|\bar{g}(t) - \bar{g}_0(t)\|_2^2 + 2\eta_k B \cdot \|\bar{g}(t) - \bar{g}_0(t)\|_2\right]}_{\textit{Error of Local Linearization}} + \underbrace{\mathbb{E}_{W,\mu}\left[\eta_k^2 \cdot \|g(t) - \bar{g}(t)\|_2^2\right]}_{\textit{Variance of Semigradient}}.
\end{aligned}$$

Proof 6.2 *For any* $W(k)(t \in [T])$*, by the convexity of* S_B *we have*

$$\begin{aligned}
\|W(k+1) - W^*\|_2^2 &= \|\Pi_{S_B}(W(k) - \eta_k \cdot z(k)) - \Pi_{S_B}\left(W^* - \eta^* \cdot \bar{z}_0^*\right)\|_2^2 \\
&\overset{(i)}{=} \|\Pi_{S_B}(W(k) - \eta_k \cdot z(k)) - \Pi_{S_B}\left(W^* - \eta_k \cdot \bar{z}_0^*\right)\|_2^2 \\
&\leq \|W(k) - W^*\|_2^2 - 2\eta_k \cdot \left(z(k) - \bar{z}_0^*\right)^\top \left(W(k) - W^*\right) + \eta_k^2 \cdot \|z(k) - \bar{z}_0^*\|_2^2.
\end{aligned} \tag{43}$$

where (i) is the result of Proposition 1. The rest of the proof mirrors the proof in Lemma 5.4 in Cai et al. (2023).

By replacing Lemma 5.4 in Cai et al. (2023) with Lemma 13, we can similarly prove the following Lemma 9 and Theorem 10.

6.3.1 Algorithm 2

We set $\eta_k = \min\{1/\sqrt{K}, \frac{1-\gamma}{8}\}$, Note that when $K \geq (8/(1-\gamma))^2$, we have $\eta_k = \frac{1}{\sqrt{K}}$. Therefore,

$$\begin{aligned}
\frac{1}{K}\sum_{k=0}^{K-1}\|\log \pi_{k+1} - \log \pi_{\theta_k}\|_\infty &\leq \frac{2}{K}\sum_{k=0}^{K-1}\left\|Q^{\text{soft}}_{r_{\theta_k},\pi_k} - Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}\right\|_\infty \\
&\leq \frac{2}{K}\sum_{k=0}^{K-1}\left\|Q^{\text{soft}}_{r_{\theta_k},\pi_k} - Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}\right\|_2 \\
&\overset{(i)}{=} \frac{2B}{K}\sum_{k=0}^{K-1}(k+2)^{-1/4} + O\left(B^{3/2}m^{-1/4} + B^{5/4}m^{-1/8}\right) \\
&\leq \frac{2B}{K}\int_{k=0}^{K-1}(k+1)^{-\frac{1}{4}} + O\left(B^{3/2}m^{-1/4} + B^{5/4}m^{-1/8}\right) \\
&\leq 2BK^{-\frac{1}{4}} + O\left(B^{3/2}m^{-1/4} + B^{5/4}m^{-1/8}\right)
\end{aligned}$$

(i) comes from the dynamic truncation design that $T = k + 2$. Therefore, we obtain that

$$\frac{1}{K}\sum_{k=0}^{K-1}\|\log \pi_{k+1} - \log \pi_{\theta_k}\|_\infty = \mathcal{O}(K^{-\frac{1}{4}}) + O\left(B^{3/2}m^{-1/4} + B^{5/4}m^{-1/8}\right) \tag{44}$$

6.3.2 Algorithm 3

Telescope (17) for k=0, . . . , K-1. When the width m goes to infinity, We have:

$$
\begin{aligned}
\frac{1}{K}\sum_{k=0}^{K-1}\left\|Q^{\text{soft}}_{r_{\theta_k},\pi_k}-Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}\right\|_2^2 &\leq 2\frac{1}{K}\sum_{k=0}^{K-1}\frac{\left[\mathbb{E}_{\text{init}}\left[\left\|W(k)-W^*_{\pi_k}\right\|_2^2\right]-\mathbb{E}_{\text{init}}\left[\left\|W(k+1)-W^*_{\pi_k}\right\|_2^2\right]+\eta_k^2\sigma_z^2\right]}{2\eta_k(1-\gamma)-8\eta_k^2} \\
&\quad + O\left(B^3m^{-1/2}+B^{5/2}m^{-1/4}\right) \\
&\overset{(i)}{\leq} 2\frac{1}{K}\sum_{k=0}^{K-1}\frac{\mathbb{E}_{\text{init}}\left[\left\|W(k)-W(k+1)\right\|_2^2\right]+\eta_k^2\sigma_z^2}{2\eta_K(1-\gamma)-8\eta_K^2} \\
&\quad + O\left(B^3m^{-1/2}+B^{5/2}m^{-1/4}\right) \\
&\overset{(ii)}{=} 2\frac{1}{K}\sum_{k=0}^{K-1}\frac{\mathbb{E}_{\text{init}}\left[\left\|\Pi_{\mathcal{F}_{B,m}}W(k)-\Pi_{\mathcal{F}_{B,m}}\tilde{W}(k+1)\right\|_2^2\right]+\eta_k^2\sigma_z^2}{2\eta_K(1-\gamma)-8\eta_K^2} \\
&\quad + O\left(B^3m^{-1/2}+B^{5/2}m^{-1/4}\right) \\
&\overset{(iii)}{\leq} 2\frac{1}{K}\sum_{k=0}^{K-1}\frac{\mathbb{E}_{\text{init}}\left[\left\|W(k)-\tilde{W}(k+1)\right\|_2^2\right]+\eta_k^2\sigma_z^2}{2\eta_K(1-\gamma)-8\eta_K^2} \\
&\quad + O\left(B^3m^{-1/2}+B^{5/2}m^{-1/4}\right) \\
&\overset{(iv)}{\leq} 2\frac{1}{K}\sum_{k=0}^{K-1}\frac{\mathbb{E}_{\text{init}}\,\eta_k^2\delta_k^2\left\|\nabla_W\widehat{Q}(s,a;W(k))\right\|_2^2+\eta_k^2\sigma_z^2}{2\eta_K(1-\gamma)-8\eta_K^2} \\
&\quad + O\left(B^3m^{-1/2}+B^{5/2}m^{-1/4}\right) \\
&\overset{(v)}{\leq} 2\frac{C_2B^2+\sigma_z^2}{K}\sum_{k=0}^{K-1}\frac{\eta_k^2}{2\eta_K(1-\gamma)-8\eta_K^2} \\
&\quad + O\left(B^3m^{-1/2}+B^{5/2}m^{-1/4}\right) \\
&\overset{(vi)}{\leq} \frac{C_3}{K}\frac{1}{2\eta_K(1-\gamma)-8\eta_K^2}+O\left(B^3m^{-1/2}+B^{5/2}m^{-1/4}\right)
\end{aligned}
\tag{45}
$$

where (i) comes from absolute inequality and the fact that the denominator is asymptotically decreasing because of our selected step length. (ii) is given by the projection step in Algorithm 3. By basic properties of the projection operator, we have (iii). (iv) is the natural result of the TD update in Algorithm 3. (v) comes from Lemma 7. (vi) can be derived by setting $C_3 = 4(C_2B^2+\sigma_z^2)$.

We select $\eta_k = \min\{k^{-\frac{3}{4}}, (1-\gamma)/8\}$. Note that when $K \geq (8/(1-\gamma))^{\frac{4}{3}}$, we have $\eta_K = K^{-\frac{3}{4}}$ and

$$K^{\frac{3}{4}}\cdot\left(2\eta_K(1-\gamma)-8\eta_K^2\right) = 2(1-\gamma)-8K^{-\frac{3}{4}} \geq 1-\gamma.$$

When $K < (8/(1-\gamma))^{\frac{4}{3}}$, we have $\eta_K = (1-\gamma)/8$ and

$$K^{\frac{3}{4}}\cdot\left(2\eta(1-\gamma)-8\eta^2\right) = K^{\frac{3}{4}}\cdot(1-\gamma)^2/8 \geq (1-\gamma)^2/8.$$

Since $|1-\gamma| < 1$, we obtain that for any $T \in \mathbb{N}$,

$$\frac{1}{K^{\frac{3}{4}}\cdot(2\eta(1-\gamma)-8\eta^2)} \leq \frac{8}{(1-\gamma)^2} \tag{46}$$

Then,

$$\begin{aligned}
\frac{1}{K}\sum_{k=0}^{K-1}\left\|\log\pi_{k+1}-\log\pi_{\theta_k}\right\|_\infty &\le \frac{2}{K}\sum_{k=0}^{K-1}\left\|Q^{\text{soft}}_{r_{\theta_k},\pi_k}-Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}\right\|_\infty \\
&\le \frac{2}{K}\sum_{k=0}^{K-1}\left\|Q^{\text{soft}}_{r_{\theta_k},\pi_k}-Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}\right\|_2 \\
&\overset{(i)}{\le} 2\sqrt{\frac{1}{K}\sum_{k=0}^{K-1}\left\|Q^{\text{soft}}_{r_{\theta_k},\pi_k}-Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}\right\|_2^2} \\
&\overset{(ii)}{\le} 2\sqrt{\frac{C_3}{K}\frac{1}{2\eta_K(1-\gamma)-8\eta_K^2}+O\left(B^3m^{-1/2}+B^{5/2}m^{-1/4}\right)} \\
&\overset{(iii)}{\le} 2\sqrt{C_3}\sqrt{\frac{1}{K^{\frac{1}{4}}}\frac{8}{(1-\gamma)^2}}+O\left(B^{3/2}m^{-1/4}+B^{5/4}m^{-1/8}\right)
\end{aligned} \tag{47}$$

where $C_3 = 2\sqrt{2C_2B^2}$. (i) comes from the Cauchy-Schwartz inequality, (ii) is due to equation (45), and (iii) is given by equation (46). Therefore, we obtain that

$$\frac{1}{K}\sum_{k=0}^{K-1}\left\|\log\pi_{k+1}-\log\pi_{\theta_k}\right\|_\infty = \mathcal{O}(K^{-\frac{1}{8}})+O\left(B^{3/2}m^{-1/4}+B^{5/4}m^{-1/8}\right) \tag{48}$$

6.4 Convergence of reward parameters

We have the following result of the objective function $L(\theta)$:

$$\begin{aligned}
L\left(\theta_{k+1}\right) &\overset{(i)}{\ge} L\left(\theta_k\right)+\left\langle\nabla L\left(\theta_k\right),\theta_{k+1}-\theta_k\right\rangle-\frac{L_c}{2}\left\|\theta_{k+1}-\theta_k\right\|^2-\mathcal{O}\left(m^{-1/2}\right) \\
&\overset{(ii)}{=} L\left(\theta_k\right)+\alpha\left(\nabla L\left(\theta_k\right),g_k\right\rangle-\frac{L_c\alpha^2}{2}\left\|g_k\right\|^2-\mathcal{O}\left(m^{-1/2}\right) \\
&= L\left(\theta_k\right)+\alpha\left(\nabla L\left(\theta_k\right),g_k-\nabla L\left(\theta_k\right)\right\rangle+\alpha\left\|\nabla L\left(\theta_k\right)\right\|^2-\frac{L_c\alpha^2}{2}\left\|g_k\right\|^2-\mathcal{O}\left(m^{-1/2}\right) \\
&\overset{(iii)}{\ge} L\left(\theta_k\right)+\alpha\left(\nabla L\left(\theta_k\right),g_k-\nabla L\left(\theta_k\right)\right\rangle+\alpha\left\|\nabla L\left(\theta_k\right)\right\|^2-2L_cL_q^2\alpha^2-\mathcal{O}\left(m^{-1/2}\right)
\end{aligned} \tag{49}$$

where (i) is from Lemma 5; (ii) follows the update formulate given in Algorithm 3 line 13; (iii) is from constant bound in Lemma 8.

Taking an expectation over both sides, it holds that

$$\begin{aligned}
&\mathbb{E}\left[L\left(\theta_{k+1}\right)\right] \\
&\ge \mathbb{E}\left[L\left(\theta_k\right)\right]+\alpha\mathbb{E}\left[\left\langle\nabla L\left(\theta_k\right),g_k-\nabla L\left(\theta_k\right)\right\rangle\right]+\alpha\mathbb{E}\left[\left\|\nabla L\left(\theta_k\right)\right\|^2\right]-2L_cL_q^2\alpha^2-\mathcal{O}\left(m^{-1/2}\right) \\
&= \mathbb{E}\left[L\left(\theta_k\right)\right]+\alpha\mathbb{E}\left[\left\langle\nabla L\left(\theta_k\right),\mathbb{E}\left[g_k-\nabla L\left(\theta_k\right)\mid\theta_k\right]\right\rangle\right]+\alpha\mathbb{E}\left[\left\|\nabla L\left(\theta_k\right)\right\|^2\right]-2L_cL_q^2\alpha^2-\mathcal{O}\left(m^{-1/2}\right) \\
&\overset{(i)}{=} \mathbb{E}\left[L\left(\theta_k\right)\right]+\alpha\mathbb{E}\left[\left\|\nabla L\left(\theta_k\right)\right\|^2\right]-2L_cL_q^2\alpha^2-\mathcal{O}\left(m^{-1/2}\right) \\
&\quad+\alpha\mathbb{E}\left[\left\langle\nabla L\left(\theta_k\right),\mathbb{E}_{\tau\sim\pi_{\theta_k}}\left[\sum_{t\ge0}\gamma^t\nabla_\theta r\left(s_t,a_t;\theta_t\right)\right]-\mathbb{E}_{\tau\sim\pi_{k+1}}\left[\sum_{t\ge0}\gamma^t\nabla_\theta r\left(s_t,a_t;\theta_t\right)\right]\right\rangle\right] \\
&\overset{(ii)}{\ge} \mathbb{E}\left[L\left(\theta_k\right)\right]-2\alpha L_q\mathbb{E}\left[\left\|\mathbb{E}_{\tau\sim\pi_{\theta_k}}\left[\sum_{t\ge0}\gamma^t\nabla_\theta r\left(s_t,a_t;\theta_k\right)\right]-\mathbb{E}_{\tau\sim\pi_{k+1}}\left[\sum_{t\ge0}\gamma^t\nabla_\theta r\left(s_t,a_t;\theta_k\right)\right]\right\|\right] \\
&\quad+\alpha\mathbb{E}\left[\left\|\nabla L\left(\theta_k\right)\right\|^2\right]-2L_cL_q^2\alpha^2-\mathcal{O}\left(m^{-1/2}\right)
\end{aligned} \tag{50}$$

$$
\begin{aligned}
&\mathbb{E}\left[\left\|\mathbb{E}_{\tau\sim\pi_{\theta_k}}\left[\sum_{t\geq 0}\gamma^t\nabla_\theta r\left(s_t,a_t;\theta_k\right)\right]-\mathbb{E}_{\tau\sim\pi_{k+1}}\left[\sum_{t\geq 0}\gamma^t\nabla_\theta r\left(s_t,a_t;\theta_k\right)\right]\right\|\right]\\
&\overset{(i)}{=}\mathbb{E}\left[\left\|\frac{1}{1-\gamma}\mathbb{E}_{(s,a)\sim d(\cdot,\cdot;\pi_{\theta_k})}\left[\nabla_\theta r\left(s,a;\theta_k\right)\right]-\frac{1}{1-\gamma}\mathbb{E}_{(s,a)\sim d(\cdot,\cdot;\pi_{k+1})}\left[\nabla_\theta r\left(s,a;\theta_k\right)\right]\right\|\right]\\
&\overset{(ii)}{\leq}\frac{2}{1-\gamma}\cdot\max_{s\in\mathcal{S},a\in\mathcal{A}}\left\|\nabla_\theta r\left(s,a;\theta_k\right)\right\|\cdot\mathbb{E}\left[\left\|d\left(\cdot,\cdot;\pi_{\theta_k}\right)-d\left(\cdot,\cdot;\pi_{k+1}\right)\right\|_{TV}\right]\\
&\overset{(iii)}{\leq}\frac{2L_r}{1-\gamma}\mathbb{E}\left[\left\|d\left(\cdot,\cdot;\pi_{\theta_k}\right)-d\left(\cdot,\cdot;\pi_{k+1}\right)\right\|_{TV}\right]\\
&\overset{(iv)}{\leq}2L_qC_d\mathbb{E}\left[\left\|Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}-Q^{\text{soft}}_{r_{\theta_k},\pi_k}\right\|\right]\\
&\overset{(v)}{\leq}2L_qC_d\sqrt{|\mathcal{S}|\cdot|\mathcal{A}|}\mathbb{E}\left[\left\|Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}-Q^{\text{soft}}_{r_{\theta_k},\pi_k}\right\|_\infty\right]
\end{aligned}
\tag{51}
$$

where (i) follows the definition $d(s,a;\pi)=(1-\gamma)\pi(a\mid s)\sum_{t\geq 0}\gamma^t\mathcal{P}^\pi\left(s_t=s\right)$; (ii) is due to distribution mismatch between two visitation measures; (iii) follows the inequality in Lemma 3; the inequality (iv) follows Lemma 6 and the fact that $\pi_{\theta_k}(\cdot\mid s)\propto\exp\left(Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}(s,\cdot)\right),\pi_{k+1}(\cdot\mid s)\propto\exp\left(Q^{\text{soft}}_{r_{\theta_k},\pi_k}(s,\cdot)\right)$ and the constant $L_q:=\frac{L_r}{1-\gamma}$; (v) follows the conversion between Frobenius norm and infinity norm. Through plugging the inequality (51) into (50), it leads to

$$
\begin{aligned}
\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\left[\left\|\nabla L\left(\theta_k\right)\right\|^2\right]\leq 2L_cL_q^2\alpha+\frac{C_1}{K}\sum_{k=0}^{K-1}\mathbb{E}\left[\left\|Q^{\text{soft}}_{r_{\theta_k},\pi_{\theta_k}}-Q^{\text{soft}}_{r_{\theta_k},\pi_k}\right\|_\infty\right]+\mathbb{E}\left[\frac{L\left(\theta_K\right)-L\left(\theta_0\right)}{K\alpha}\right]\\
+\mathcal{O}\left(m^{-1/2}\right)
\end{aligned}
\tag{52}
$$

6.4.1 Theorem 1

Rearranging the inequality above and denote $C_1:=4C_dL_q^2\sqrt{|\mathcal{S}|\cdot|\mathcal{A}|}$, it holds that

$$
\begin{aligned}
\frac{1}{K}\sum_{K=0}^{K-1}\mathbb{E}\left[\left\|\nabla L\left(\theta_K\right)\right\|^2\right]=\mathcal{O}\left(K^{-\sigma}\right)+\mathcal{O}\left(K^{-1+\sigma}\right)+\mathcal{O}\left(K^{-\frac{1}{4}}\right)\\
+O\left(m^{-1/2}+B^{3/2}m^{-1/4}+B^{5/4}m^{-1/8}\right)
\end{aligned}
\tag{53}
$$

6.4.2 Theorem 2

Rearranging the inequality above and denote $C_1:=4C_dL_q^2\sqrt{|\mathcal{S}|\cdot|\mathcal{A}|}$, it holds that

$$
\begin{aligned}
\frac{1}{K}\sum_{K=0}^{K-1}\mathbb{E}\left[\left\|\nabla L\left(\theta_K\right)\right\|^2\right]=\mathcal{O}\left(K^{-\sigma}\right)+\mathcal{O}\left(K^{-1+\sigma}\right)+\mathcal{O}\left(K^{-\frac{1}{8}}\right)+\mathcal{O}\left(K^{-1+\sigma}\right)+\mathcal{O}\left(K^{-\frac{1}{4}}\right)\\
+O\left(m^{-1/2}+B^{3/2}m^{-1/4}+B^{5/4}m^{-1/8}\right)
\end{aligned}
\tag{54}
$$

6.5 Global Optimality of reward and policy parameters

Since $L(\theta)$ is nonconcave in θ, showing the global optimality of (3) is non-trivial. Now we consider the general case that reward is parametrized by neural networks, which means that the stationary point

of $L(\theta)$ is not necessarily the global optimum. Therefore, we translate this problem to an equivalent saddle point problem

$$\max_{\theta}\min_{\pi} L(\theta,\pi) := \mathbb{E}_{\tau^{\mathrm{E}}\sim(\eta,\pi^{\mathrm{E}},P)}\left[\sum_{t=0}^{\infty}\gamma^t\left(r\left(s_t,a_t;\theta\right)\right)\right] - \mathbb{E}_{\tau^{\mathrm{A}}\sim(\eta,\pi,\widehat{P})}\left[\sum_{t=0}^{\infty}\gamma^t\left(r\left(s_t,a_t;\theta\right)+\mathcal{H}\left(\pi\left(\cdot\mid s_t\right)\right)\right)\right]. \tag{55}$$

Lemma 14 *It holds for any $\theta'\in S_{B_\theta}$ that*

$$\mathbb{E}_{init}\left[L\left(\pi_{\theta_k},\theta'\right)-L\left(\pi_{\theta_k},\theta_k\right)\right] = \mathbb{E}_{init}\left[\nabla_\theta L\left(\pi_{\theta_k},\theta_k\right)^\top\left(\theta'_k-\theta_k\right)\right] + \mathcal{O}\left(B^{3/2}\cdot m^{-1/4}\right) \tag{56}$$

Proof 6.3 $\left[\phi_{\theta,b}(s,a)\right]_l = m^{-1/2}\cdot b_l\cdot \mathbb{1}\left\{(s,a)^\top[\theta]_l>0\right\}\cdot(s,a)$

$$\begin{aligned}
&L\left(\pi_{\theta_k},\theta'\right)-L\left(\pi_{\theta_k},\theta_k\right) = \left\langle r'-r_k,\nu_{\mathrm{E}}-\nu_k\right\rangle_{\mathcal{S}\times\mathcal{A}}\\
&\quad=\left(\left\langle\phi_{\theta_k}^\top\left(\theta'-\theta_k\right),\nu_{\mathrm{E}}-\nu_k\right\rangle_{\mathcal{S}\times\mathcal{A}}+\left\langle\phi_{\theta'}^\top\theta'-\phi_{\theta_k}^\top\theta',\nu_{\mathrm{E}}-\nu_k\right\rangle_{\mathcal{S}\times\mathcal{A}}\right)\\
&\quad\le\left(\theta'-\theta_k\right)^\top\nabla_\theta L\left(\pi_{\theta_k},\theta_k\right)+(1-\gamma)^{-1}\cdot\left(\left\|\phi_{\theta_k}^\top\theta'-\phi_{\theta'}^\top\theta'\right\|_{1,\nu_k}+\left\|\phi_{\theta_k}^\top\theta'-\phi_{\theta'}^\top\theta'\right\|_{1,\nu_{\mathrm{E}}}\right),
\end{aligned} \tag{57}$$

Then we have

$$\begin{aligned}
&\mathbb{E}_{init}\left[L\left(\pi_{\theta_k},\theta'\right)-L\left(\pi_{\theta_k},\theta_k\right)\right]\\
&\quad\le\mathbb{E}_{init}\left[\left(\theta'-\theta_k\right)^\top\nabla_\theta L\left(\pi_{\theta_k},\theta_k\right)+(1-\gamma)^{-1}\cdot\left(\left\|\phi_{\theta_k}^\top\theta'-\phi_{\theta'}^\top\theta'\right\|_{1,\nu_k}+\left\|\phi_{\theta_k}^\top\theta'-\phi_{\theta'}^\top\theta'\right\|_{1,\nu_{\mathrm{E}}}\right)\right]\\
&\quad\le\mathbb{E}_{init}\left[\left(\theta'-\theta_k\right)^\top\nabla_\theta L\left(\pi_{\theta_k},\theta_k\right)\right]+\mathcal{O}\left(B^{3/2}\cdot m^{-1/4}\right),
\end{aligned} \tag{58}$$

From Lemma 14, we can see that L is approximately concave.
Recall that a tuple $\left(\tilde\theta,\pi_{\tilde\theta}\right)$ is called a saddle point of $L(\cdot,\cdot)$ if the following condition holds:

$$L\left(\theta,\pi_{\tilde\theta}\right)\le L\left(\tilde\theta,\pi_{\tilde\theta}\right)\le L(\tilde\theta,\pi) \tag{59}$$

for any other reward parameter θ and policy π. To show that $\left(\tilde\theta,\pi_{\tilde\theta}\right)$ satisfies the condition (59), we prove the following conditions respectively:

$$\tilde\theta\in\arg\max_{\theta} L\left(\theta,\pi_{\tilde\theta}\right), \tag{60}$$

$$\pi_{\tilde\theta}\in\arg\min_{\pi} L(\tilde\theta,\pi). \tag{61}$$

Here, we first show that any stationary point $\tilde\theta$ of the objective $L(\cdot)$ satisfies the optimality condition (60). Recall the gradient expression of $L(\cdot)$, any stationary point $\tilde\theta$ of the objective $L(\cdot)$ satisfies the following first-order condition:

$$\nabla L(\tilde\theta)=\mathbb{E}_{\tau\sim\mathcal{D}}\left[\sum_{t\ge0}\gamma^t\nabla_\theta r\left(s_t,a_t;\tilde\theta\right)\right]-\mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t\ge0}\gamma^t\nabla_\theta r\left(s_t,a_t;\tilde\theta\right)\right]=0 \tag{62}$$

Then we can go back to the formulation $L(\cdot,\cdot)$ in (23). Given any fixed policy π, we can show the gradient of $L(\pi,\theta)$ w.r.t. the reward parameter θ as below:

$$\nabla_\theta L(\pi_\theta,\theta)=\mathbb{E}_{\tau\sim\mathcal{D}}\left[\sum_{t\ge0}\gamma^t\nabla_\theta r\left(s_t,a_t;\tilde\theta\right)\right]-\mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t\ge0}\gamma^t\nabla_\theta r\left(s_t,a_t;\tilde\theta\right)\right] \tag{63}$$

Due to the first-order condition we show in (62), we obtain the following result:

$$\nabla_\theta L\left(\theta = \tilde{\theta}, \pi = \pi_{\tilde{\theta}}\right) = 0. \tag{64}$$

Recall that we have shown $L(\cdot,\cdot)$ is concave in terms of the reward parameter θ given any fixed policy π, if the neural network is sufficiently large. Due to the concavity as shown in 14 and the condition in (64), we have completed the proof of the optimality condition (60).

Then we prove the optimality condition (40). Recall that $\pi_{\tilde{\theta}}$ is the optimal policy defined in (3) under the reward parameter θ. By observing the objective $L(\cdot,\cdot)$, we obtain the following result:

$$\begin{aligned} L(\theta,\pi) := & \underbrace{\mathbb{E}_{\tau^{\mathrm{E}}\sim(\eta,\pi^{\mathrm{E}},P)}\left[\sum_{t=0}^{\infty}\gamma^t\left(r\left(s_t,a_t;\theta\right)\right)\right]}_{\text{Term I I}_1\text{ independent of }\pi} \\ & \underbrace{-\,-\mathbb{E}_{\tau^{\mathrm{A}}\sim(\eta,\pi,\widehat{P})}\left[\sum_{t=0}^{\infty}\gamma^t\left(r\left(s_t,a_t;\theta\right)+\mathcal{H}\left(\pi\left(\cdot\mid s_t\right)\right)\right)\right]}_{\text{Term I}_2\text{ : a function of the policy }\pi}. \end{aligned} \tag{65}$$

Here, we can observe that the second term in (65) is exactly the objective in (3). Moreover, since $\pi_{\tilde{\theta}}$ is the optimal policy defined in (3) under reward parameter $\tilde{\theta}$, we obtain the result that

$$\pi_{\tilde{\theta}} \in \arg\min_{\pi} L(\tilde{\theta},\pi) \tag{66}$$

which completes the proof to show the optimality condition (61). Hence, we obtain that any stationary point $\tilde{\theta}$ of the surrogate objective $L(\cdot,\cdot)$ together with its optimal policy $\pi_{\tilde{\theta}}$ is a saddle point of $L(\cdot,\cdot)$

Since L is concave in the reward parameter under over-parametrized case, we have

$$\tilde{\theta} \in \arg\max_{\theta}\min_{\pi_\theta} L(\theta,\pi) = \arg\max_{\theta} L(\theta). \tag{67}$$

we obtain that for any saddle point $\left(\tilde{\theta},\pi_{\tilde{\theta}}\right)$ of $L(\cdot,\cdot)$, so the reward parameter $\tilde{\theta}$ is a globally optimal solution of the objective $L(\cdot)$ in (3).